Identity as Event History

Irreversibility, Commitment, and the Ontology of Irreversible Processes

Flyxion

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Abstract

Spherepop is a process-oriented formalism in which the identity of an object is constituted by the irreversible sequence of transformations that produced it rather than by its present configuration. This essay develops the framework through a single sustained argument in two movements, unified by the central claim that identity as event history is not a primitive but a derived notion: it is what remains when possibility has been progressively collapsed.

The first movement establishes the formal infrastructure. Event words and normal forms provide a canonical symbolic encoding of object identity. Directed acyclic event graphs and causal posets capture branching and parallel structure. A categorical semantics treats histories as functors from causal index categories, and merge events are characterised as pushout constructions. Operational semantics, historical complexity measures, the reconstruction problem, and a stratified family of equivalence relations complete the treatment. The first movement culminates in the Necessity Theorem: under genuine irreversibility, extensional identity fails as a faithful criterion in both directions, and historical identity is the only consistent alternative.

The second movement reveals the preimage of the first. Event histories arise from a deeper dynamical process: the monotone contraction of a space of possibilities. The Spherepop state at any stage is a pair (Ω_t, H_t) of a shrinking option space and a growing commitment history. The event word is the image of the history projection Π_{hist} applied to this deeper dynamics, making identity a quotient of an irreversible process rather than a primitive object. Entropy is derived directly from the geometry of option-space contraction. Operators are

reinterpreted as typed generators of event words, classifying the four fundamental modes of exclusion. Responsibility weights measure the provenance of each contraction, running as a parallel scalar beside the entropy measure.

The two movements are unified by a categorical diagram relating the trajectory category $\mathcal{T}$, the Free History Category $\mathbb{P}(\Omega)$, and the category of observable states $\mathcal{S}$, with projections Π_{hist} and **Collapse** connecting the three levels and neither admitting a right inverse. The Worldhood Theorem characterises when a system genuinely inhabits this structure—when it performs irreversible commitments for which it bears non-zero responsibility—as opposed to simulating it from a fixed option space. A discrete variational principle then provides the generative law: realised trajectories extremize an action functional that balances entropy, responsibility, and coherence, giving the history of an object the status of an optimal trajectory, not merely a recorded one. The essay concludes by placing this framework in contrast with the set-theoretic ontology it replaces: from membership to commitment, from static equivalence to irreversible construction, from objects defined by their contents to processes defined by what they have made impossible.

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1 Introduction

A recurring question in the foundations of mathematics and formal ontology concerns the correct criterion of identity for objects that exist within processes rather than in isolation. Classical set theory locates identity in the membership relation: two sets are equal precisely when they contain exactly the same elements, irrespective of how those sets came to exist or in what order their elements were assembled. This extensional criterion is powerful and convenient, but it discards what may be the most significant feature of many objects encountered in physics, computation, and the theory of complex systems: that an object's identity is inseparable from the history that generated it.

Spherepop is a formalism developed to address this gap directly. Its central claim is that objects should be represented not as static containers but as records of transformations, and that identity should be adjudicated by comparing those records rather than by comparing present states. The formalism takes irreversibility as its organising principle. Because the transformations that produce objects cannot in general be undone, the historical record accumulates monotonically; each new event appends to a trace that cannot be retracted. Two objects are identical in the Spherepop sense if and only if they are the products of identical histories.

This essay develops the framework through a single sustained argument in two interlocking movements, unified by the insight that identity as event history is not primitive but derived: it is what remains when possibility has been progressively collapsed.

The first movement takes identity as event history as its immediate object of study. It establishes the canonical symbolic encoding of object identity through event words and normal forms, develops the directed acyclic graph and causal poset representations of branching histories, provides a categorical semantics in which histories are functors from causal index categories, characterises merge events as pushout constructions, analyses operational semantics, historical complexity measures, and the reconstruction problem, and situates the framework within the theory of free traced monoidal categories and provenance systems. The culminating formal result of this movement is the Necessity Theorem (Section 17): under conditions of genuine irreversibility, extensional identity fails as a faithful criterion in both directions, and historical identity is the only consistent alternative. This is not the claim that historical identity is richer or more expressive—which is obvious—but the stronger claim that extensional identity is positively inadequate: it conflates objects that are genuinely distinct and fails to

cohere with the structure that irreversibility imposes.

The second movement reveals the preimage of the first. It asks: what is the structure from which event histories arise? The answer is that an event history is not primitive but is the irreversible boundary trace of a deeper process: the monotone contraction of a space of possibilities. A Spherepop object in the process of being formed is a state pair (Ω_t, H_t) consisting of what has already been irrevocably committed and what still remains possible. The option space Ω_t shrinks as the commitment history H_t grows, and the event word that the first movement treats as primary is the image of the history projection Π_{hist} applied to this deeper dynamics. This reinterpretation makes identity a quotient of an irreversible process: two objects are identical when their trajectories through possibility space produce the same ordered boundary trace, regardless of how their option spaces evolved internally along the way.

The correct relationship between the two movements is vertical, not lateral. The second does not extend the first by appending new content; it deepens the first by revealing what the first presupposes. Identity as event history is the result of projecting the possibility-history dynamics onto its history coordinate. Every formal structure in the first movement has a counterpart at this deeper level, and once the connection is established the two movements read as one: what identity is, and how identity comes into existence at all.

The unification is expressed formally in a categorical diagram (Section 24) relating three levels of description—the trajectory category $\mathcal{T}$, the Free History Category $\mathbb{P}(\Omega)$, and the category of observable states $\mathcal{S}$—connected by the projections Π_{hist} and **Collapse**, neither of which admits a right inverse. Information flows in one direction only: from the full geometry of possibility, through the ordered record of commitment, to the observable shadow of present state. No functor reverses any of these projections. This is not a technical inconvenience but the formal expression of irreversibility at every level of the ontology.

The Worldhood Theorem (Section 25) then characterises, within this structure, the distinction between systems that genuinely inhabit it and systems that merely simulate its surface. A system that does not contract its option space cannot accumulate a meaningful history and therefore cannot possess identity in the relevant sense. It may produce outputs indistinguishable from those of a world-inhabiting system, but it does so from an unchanged space of possibilities, and the distinction is formal, not merely

descriptive.

The variational principle of Section 28 completes the framework by providing a generative law. Realised trajectories extremize an action functional that balances three competing quantities: entropy, measuring the magnitude of option-space contraction; responsibility, measuring its provenance; and coherence, measuring its structural compatibility with the remaining possibilities. The histories that Spherepop takes as the bearers of identity are therefore not arbitrary sequences but optimal ones: traces of trajectories that could not have been otherwise without increasing the total cost of their construction.

This orientation connects Spherepop to a broad family of existing frameworks. In category theory, the primacy of morphisms over objects is a foundational methodological commitment [Mac Lane, 1998]; histories composed of irreversible events are naturally interpreted as compositions of morphisms, and the event histories of objects become diagrams in causal categories. In theoretical computer science, the notion that computational identity should be grounded in execution traces rather than in present memory states appears implicitly throughout operational semantics [Abramsky and Jung, 1994]. In concurrency theory, Mazurkiewicz traces [Mazurkiewicz, 1987] provide a precedent for treating event sequences as equivalence classes modulo independent commutation. Landauer's principle [Landauer, 1961] establishes the physical lower bound on the cost of information destruction that the entropy-from-contraction framework makes precise. And Jonas's argument [Jonas, 1979] that irreversibility entails responsibility provides the warrant for the responsibility-weighted histories that the second movement introduces.

The essay proceeds as follows. Section 2 presents the worked example. Section 3 formalises event words and normal forms. Section 4 introduces the state pair, the three axioms, and the reinterpretation of event words as boundary traces of possibility-space dynamics. Section 5 derives entropy from the contraction of Ω_t. Sections 6 through 20 develop the full formal infrastructure of the first movement: categorical interpretation, DAGs, normalization, functorial semantics, pushouts, entropy landscapes, partial orders, complexity measures, operational semantics, reconstruction, and provenance. Section 16 reformulates identity as projected history. Section 17 proves the Necessity Theorem. Section 18 develops the stratified equivalence lattice. Section 19 situates the framework within free traced monoidal categories. Section 21 introduces operators

as typed generators of event words. Section 22 develops commitment mechanics. Section 23 formalises ethical non-neutrality. Section 24 establishes the global categorical diagram. Section 25 states and proves the Worldhood Theorem. Section 26 examines large language models as a boundary case. Section 27 specifies the stratified architecture. Section 28 derives the variational principle for commitment. Section 29 concludes by placing the framework in contrast with the set-theoretic ontology it replaces. Appendices treat the BNF grammar, rewriting rules, connections to trace theory, Petri nets, string diagrams, and derived causal categories.

2 A Worked Example

To make the core principle concrete, consider an initial region R and a sequence of three events. The first event, E_1, causes R to split into two descendant regions A and B. The second event, E_2, is a transformation internal to A that produces a new region C. The third event, E_3, causes the regions B and C to merge, producing the region D. Although the informal description is clear, the formal representation of this history must record not merely which events occurred but in which order they occurred and from which states each event proceeded.

Using the normal-form notation to be fully developed in Section 3, the history of each region is recorded as follows. Both A and B arose directly from R through the single event E_1, so their normal forms are identical:

$$(\text{spherepop}(A)) = (E_1), \qquad (\text{spherepop}(B)) = (E_1).$$

The region C, however, arose from A through the subsequent event E_2, so its history includes both events:

$$(\text{spherepop}(C)) = (E_1, E_2).$$

Finally, D arose through the merger of B and C, which itself required all three events in order, so:

$$(\text{spherepop}(D)) = (E_1, E_2, E_3).$$

The identity of each object is thus entirely determined by the ordered sequence of events that produced it. This is already a departure from structural or compositional criteria: a criterion based on present structure would describe D solely in terms of what

D currently contains, with no record of whether D arose from a merger of previously split regions or from some entirely different sequence of transformations.

To make this distinction sharp, consider a second process. Suppose again that E_1 causes R to split into A and B, but now suppose that instead of E_2 there occurs a different event E_3' that directly merges A and B into a region D', without the intermediate transformation that in the first process generated C. The normal form of D' is then:

$$(\text{spherepop}(D')) = (E_1, E_3').$$

Comparing the normal forms of D and D',

$$(\text{spherepop}(D)) = (E_1, E_2, E_3), \qquad (\text{spherepop}(D')) = (E_1, E_3'),$$

the two histories are evidently distinct: D accumulated three events while D' accumulated only two, and the events themselves differ. The Spherepop identity criterion therefore concludes that D and D' are not identical objects, even if their present structural descriptions happen to agree.

This is the core claim of the formalism. Structural indistinguishability is insufficient for identity. Two objects are identical in Spherepop precisely when their complete generative histories, recorded in normal form, are identical.

3 Event Words and Normal Forms

The worked example of Section 2 illustrates the central representational idea intuitively. To deploy it rigorously, the notion of a history must be formalized as a structured symbolic object.

Let $\mathcal{E}$ denote the set of possible events in a given system. Each event in $\mathcal{E}$ corresponds to an irreversible transformation that either produces new regions, modifies existing ones, or both. The precise physical or computational content of these events may vary; what matters for the formalism is that they are discrete, ordered, and irreversible.

Definition 3.1 (Event word)**.** An *event word* over $\mathcal{E}$ is a finite ordered sequence

$$w = (E_{i_1}, E_{i_2}, \ldots, E_{i_k})$$

where each $E_{i_j} \in \mathcal{E}$. The length of w is $|w| = k$. The empty event word, denoting an object with no generative history, is written ε.

Because the events constituting a word are ordered, the same multiset of events occurring in different sequences represents a different history. This ordering is not arbitrary: it encodes the causal and temporal precedence relations among the events, about which more will be said when branching histories are considered.

Definition 3.2 (Spherepop normal form)**.** Spherepop associates to every object X a canonical expression, the *normal form* of X, written $(\text{spherepop}(X))$, which is defined to be the event word representing the complete generative history of X:

$$(\text{spherepop}(X)) = (E_{i_1}, E_{i_2}, \ldots, E_{i_k}).$$

The normal form is the primary identity criterion of the formalism. Two objects are identical if and only if their normal forms coincide:

$$X \equiv Y \quad \Longleftrightarrow \quad (\text{spherepop}(X)) = (\text{spherepop}(Y)).$$

This principle replaces identity criteria grounded in structural or compositional properties. Objects that are structurally indistinguishable may nevertheless be distinct if their histories differ, and objects that arose through entirely different structural pathways may be identical if those pathways produced identical event sequences.

It is useful to visualize this criterion geometrically. Figure 1 represents an event word as a directed trace along a time axis, with the irreversible events E_1, E_2, E_3 appearing as marked transitions separating successive phases of the history. The tick marks crossing the trace axis emphasize that each transition is a genuine discontinuity: no reversion to a prior state is possible. The brace beneath the axis indicates that the normal form $(\text{spherepop}(D))_{\text{nf}} = (E_1, E_2, E_3)$ records the complete ordered content of that trace.

The geometric picture reinforces a fundamental intuition: an object's position on this trace cannot be recovered by inspecting the object alone. Just as a physical process that has dissipated energy cannot be run backwards to recover the initial configuration, a Spherepop object does not carry within itself the information necessary to reconstruct its generating history. The normal form is an external record, not an intrinsic property.

This historical criterion avoids several classical difficulties of set-based ontology. Ob-

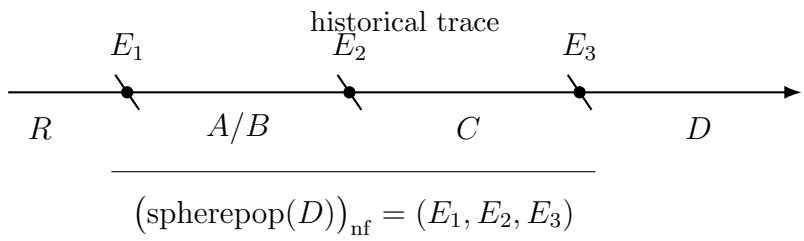

Figure 1: A geometric representation of an event word as an irreversible trace. Tick marks indicate irreversible transitions; the brace records the corresponding normal form.

jects in Spherepop are not defined by membership in collections but by the irreversible sequences that produced them. The ontology is therefore inherently process-based: objects are histories rather than containers, and the content of an object is its temporal structure rather than its spatial extension.

4 Possibility as the Preimage of History

The event-word formalism of Section 3 provides a complete and canonical representation of object identity in terms of irreversible sequences of transformations. Each object X is identified with a normal form

$$(\text{spherepop}(X)) = (E_{i_1}, E_{i_2}, \ldots, E_{i_k}),$$

and identity is adjudicated by equality of these sequences. This construction is sufficient for distinguishing objects once their histories are given. But it leaves a prior question unanswered: what is the structure from which such histories arise? To ask what an event word records is already to presuppose that it records something rather than being primitive. This section makes the answer precise.

The central claim is that an event history should be understood not as a primitive object but as the irreversible boundary trace of a deeper process, namely the progressive elimination of possibilities. The event word is not merely a record of what happened; it is the ordered residue of what has been made impossible. Identity as event history is therefore a derived notion: it is what remains when possibility has been progressively collapsed.

4.1 The State Pair

At any stage of a process we do not have merely a partial history, but a pair consisting of what has already been irrevocably committed and what still remains open. This motivates the following definition.

Definition 4.1 (Spherepop state)**.** A *Spherepop state* at stage t is a pair

$$(\Omega_t, H_t),$$

where Ω_t is a finite set of admissible events representing the residual option space, and $H_t \in \mathcal{E}^*$ is the event word representing the committed history accumulated up to stage t.

The two components play fundamentally different structural roles. The history H_t is an ordered, append-only record of irreversible commitments; it carries the full ordering structure established in Section 3 and required by the DAG construction to follow in Section 7. The option space Ω_t is an unordered set, because the future has not yet been differentiated by the act of choice: the possibilities that remain do not yet stand in any causal relation to each other, and imposing an order on them would be to read the future backwards from the past.

This asymmetry between the ordered history and the unordered option space is not a notational convention; it is the formal expression of the asymmetry of time. The past has a definite sequence because it is constituted by the specific order in which exclusions occurred. The future is an unordered cloud because no commitment has yet been made that would differentiate one remaining possibility from another.

4.2 The Axioms of Irreversible Evolution

Three axioms govern the evolution of the state pair. They are not independent stipulations but consequences of taking irreversibility seriously as the foundational property of Spherepop processes.

Axiom 4.2 (Irreversibility)**.** For any admissible transition from stage t to stage $t+1$, the option space satisfies $\Omega_{t+1} \subseteq \Omega_t$ and the commitment history satisfies $H_{t+1} = H_t \cdot w$ for some non-empty extension $w \in \mathcal{E}^*$. No operation within the formalism can enlarge Ω_t or remove any element from H_t.

Axiom 4.3 (Exclusion Primitivity)**.** The only admissible operation on Ω_t is the re-

moval of elements. An event that has been excluded from the option space does not return. Exclusion is primitive; there is no complementary inclusion operation.

Axiom 4.4 (History Sufficiency)**.** Every observable property of the process at stage t factors through the commitment history H_t together with a fixed initial state s_0. No observable property depends on Ω_t independently of the history that produced the current option space by successive exclusion from Ω_0.

Axiom 4.2 encodes the directed acyclic structure that will be made precise in Section 7: because the option space can only shrink and the history can only grow, the evolution of the state pair traces a path through a strictly ordered sequence of states, with no cycles and no backward steps. Axiom 4.3 ensures that the boundary of the excluded region grows monotonically; combined with Axiom 4.2, it implies that every event word is a record of irreversible decisions. Axiom 4.4 ensures that the pair (Ω_t, H_t) is not merely a bookkeeping device but carries the full semantic content of the process: state is reconstructible from history, and the option space is reconstructible from the history together with knowledge of what was initially available.

4.3 Event Words as Boundary Traces

With the state pair in hand, we can now give a precise account of what the Spherepop normal form records.

Proposition 4.5 (History as projection of possibility dynamics)**.** *Let (Ω_0, ε) be an initial state and let*

$$(\Omega_0, \varepsilon) \longrightarrow (\Omega_1, H_1) \longrightarrow \cdots \longrightarrow (\Omega_k, H_k)$$

be a sequence of irreversible transitions satisfying Axiom 4.2. Then the event word $H_k = (E_{i_1}, \ldots, E_{i_k})$ is the ordered record of the successive contractions

$$\Omega_0 \supseteq \Omega_1 \supseteq \cdots \supseteq \Omega_k,$$

and the normal form $(\mathrm{spherepop}(X))_{\mathrm{nf}} = H_k$ *encodes the trajectory of the option space under monotone exclusion.*

Proof. Each transition appends exactly one event to the history and removes at least that event from the option space by Axiom 4.2 and Axiom 4.3. The sequence of

appended events is precisely $(E_{i_1}, \ldots, E_{i_k})$ and the sequence of option spaces is $\Omega_0 \supseteq \cdots \supseteq \Omega_k$. The event word is therefore a complete record of the exclusion sequence. $\square$

The normal form (spherepop(X)) is in this sense a boundary object: it records the sequence of exclusions but not the internal structure of the option space at each stage. Two distinct trajectories of Ω_t may produce identical event words if they eliminate the same events in the same order, but the event word itself contains no explicit representation of the alternatives that were available but not taken. This is precisely the information that is lost in the projection from full dynamics to boundary trace, and it is the formal content of the historical reconstruction problem taken up in Section 15.

4.4 Identity as Projection

The reformulation of history as a projection from possibility dynamics allows us to factor the construction of identity into two stages. The first stage is dynamical: a process evolves within the space of possibility-history pairs, generating a trajectory through states of diminishing possibility and growing commitment. The second stage is projective: the identity of the resulting object is obtained by discarding the option space and retaining only the accumulated commitment history.

Definition 4.6 (History projection)**.** The *history projection* is the map

$$\Pi_{\text{hist}} : (\Omega_t, H_t) \longmapsto H_t$$

which discards all information about the residual option space and retains only the ordered commitment history.

The Spherepop identity criterion is then a statement about the image of this projection: two objects are identical precisely when their histories coincide under Π_{hist}, that is, when the ordered record of their successive exclusions is the same. Different trajectories through the space of possibility-history pairs that happen to produce the same sequence of exclusions yield the same Spherepop identity, because Π_{hist} maps them to the same image. This is not an approximation or a convention; it is the formal sense in which history is what matters and residual possibility is not part of the identity of a completed object.

4.5 Consequence: Irreversibility as Definitional

The key insight that this reformulation yields is that irreversibility is not an external constraint imposed on the formalism but a definitional feature of the projection Π_{hist}. An event word grows by appending; it does not shrink. An option space shrinks by exclusion; it does not grow. The asymmetry between them is built into the definition of the state pair and the axioms that govern it. When we say that Spherepop objects have irreversible histories, we are saying that history is the image of a fundamentally directional process, one that moves from larger to smaller option spaces and from shorter to longer commitment records, and that the arrow of this process is not reversible because there is no operation in the formalism that takes a smaller option space back to a larger one or a longer history back to a shorter one.

The event-word formalism of Section 3, the DAG structure of Section 7, and the normalization theory of Section 8 all describe properties of the image of Π_{hist}. What the present section establishes is the preimage: the space of possibility-history dynamics from which those images are derived. Every subsequent formal development in this essay can be read in two registers simultaneously—as a statement about event words, which is the register of the boundary trace, and as a statement about the trajectories in (Ω_t, H_t) space that produced those event words.

5 Entropy as Contraction of Possibility

The previous section reinterpreted event histories as boundary traces of a deeper dynamical process: the progressive contraction of a space of possibilities. We now derive the entropy measure of a history directly from this contraction. Rather than introducing entropy as an external quantity assigned to events by a weight function, we show that it emerges naturally from the geometry of option spaces under successive exclusion.

5.1 Entropy as Logarithmic Volume of Possibility

Let (Ω_0, ε) be an initial state and let Ω_t denote the residual option space at stage t. We define the possibility-entropy at stage t by

$$S_t \;=\; \log|\Omega_t|,$$

where $|\Omega_t|$ is the cardinality of the remaining possibilities. If Ω_t carries a probability measure μ reflecting the relative accessibility of remaining events, the definition extends to $S_t = \log \mu(\Omega_t)$, but for most purposes the cardinality version suffices.

This definition captures the intuitive idea that entropy measures the size of the future. A large option space corresponds to high possibility-entropy: many distinct continuations remain open. A small option space corresponds to low possibility-entropy: the future has become highly constrained by the commitments already made.

5.2 Irreversibility as Entropy Descent

Under Axiom 4.2, the option space shrinks monotonically: $\Omega_{t+1} \subseteq \Omega_t$. It follows immediately that

$$S_{t+1} \;=\; \log|\Omega_{t+1}| \;\leq\; \log|\Omega_t| \;=\; S_t.$$

Possibility-entropy is non-increasing along every admissible trajectory. This is not a contingent fact about particular systems but a direct consequence of the axioms.

This relationship to classical entropy deserves comment. In thermodynamics, entropy increases along irreversible processes: information about the detailed microstate is dispersed into inaccessible degrees of freedom. Here, entropy decreases along irreversible processes: accessible alternatives are progressively eliminated. The two are dual descriptions of the same underlying directionality. What is lost from the option space is exactly what is dispersed into the irrecoverable past. The event word records the eliminations; the entropy measures their cumulative magnitude.

5.3 Event Contributions as Entropy Drops

Each transition contributes a discrete reduction in possibility-entropy. For a transition $(\Omega_t, H_t) \to (\Omega_{t+1}, H_{t+1})$, the entropy drop is

$$\Delta S_t \;=\; S_{t+1} - S_t \;=\; \log\frac{|\Omega_{t+1}|}{|\Omega_t|} \;\leq\; 0.$$

We associate to each event E_{i_j} appended at step j a natural weight derived from the option space dynamics:

$$w(E_{i_j}) \;=\; -\Delta S_{j-1} \;=\; \log\frac{|\Omega_{j-1}|}{|\Omega_j|}.$$

This weight measures how many possibilities are eliminated when the event occurs. An event that reduces a large option space by a single element carries small weight; an event that halves the option space carries weight $\log 2$.

Summing over a complete history $H_k = (E_{i_1}, \ldots, E_{i_k})$:

$$\sum_{j=1}^{k} w(E_{i_j}) \;=\; \sum_{j=1}^{k} \log \frac{|\Omega_{j-1}|}{|\Omega_j|} \;=\; \log \frac{|\Omega_0|}{|\Omega_k|}.$$

The total historical weight is exactly the log-ratio of the initial to the terminal option space: the total contraction of possibility that the history represents. The event word does not merely record what happened; it encodes the cumulative loss of possibility required to produce the object.

5.4 Historical Complexity Derived from Contraction

The definition of historical complexity can now be given a precise geometric form. For an object X arising from a trajectory with initial state Ω_0 and terminal option space Ω_k, define

$$S(X) \;=\; \log \frac{|\Omega_0|}{|\Omega_k|}.$$

This is the logarithm of the ratio by which the space of possibilities has been contracted in order to produce X. Objects with high $S(X)$ arose from trajectories that carved deeply into the option space, leaving behind a narrow corridor of remaining alternatives. Objects with low $S(X)$ arose from histories that barely constrained the future.

This definition subsumes the earlier formulation of complexity as a weighted event count. The event-count formulation, which assigned an external weight $w : \mathcal{E} \to \mathbb{R}_{\geq 0}$ to each event type and summed these weights over the history, is recovered as a special case by choosing $w(E)$ to approximate the entropy drop $-\Delta S$ associated with each event type. The present formulation is more fundamental because it derives the weights from the option space dynamics rather than assigning them externally.

5.5 Geometric Interpretation: Widening Trace as Illusion

The earlier geometric picture of historical complexity as a widening trace (described further in Section 11) can now be understood more precisely. The apparent widening

of the trace as events accumulate reflects the growing informational content of the commitment history: each additional event adds information to the record of what has been excluded. But from the perspective of the option space, the picture is exactly reversed: the option space narrows as the history widens.

The widening trace and the narrowing option space are dual representations of the same process, viewed from opposite ends of the projection Π_{hist} introduced in Section 4. The widening trace is the image under that projection; the narrowing option space is the domain. The apparent visual metaphor of growth is therefore a representation of the history side of the dynamics; the underlying reality of the possibility side is one of contraction.

5.6 Toward Responsibility Weights

The entropy formulation treats all eliminations as quantitatively comparable, differing only in magnitude. But events differ not only in how much possibility they eliminate, but in the character of that elimination: whether it arises from structural constraint, from deliberate commitment, or from principled refusal. This distinction cannot be captured by the entropy weight $w(E)$ alone, because that weight is defined entirely by the cardinality change in Ω_t and is indifferent to the provenance of the exclusion.

To capture this distinction, we introduce a refinement of the weight structure. Each event E in a history carries not only an entropy weight $w(E)$ derived from the contraction of the option space, but a responsibility weight $\rho(E)$ that records the degree to which the elimination of possibilities is attributable to the system's own agency. The total responsibility of an object is then

$$\mathcal{R}(X) \;=\; \sum_{j=1}^{k} \rho(E_{i_j}),$$

which provides a second independent measure on histories running in parallel with the entropy measure $S(X)$. While $S(X)$ measures how much possibility has been contracted, $\mathcal{R}(X)$ measures who or what is answerable for that contraction.

Both measures arise from the same underlying structure—the irreversible contraction of Ω_t—but they answer different questions about that structure. One asks about magnitude; the other asks about provenance. The full elaboration of the responsibility

measure, including the operator vocabulary that distinguishes the types of exclusion from which it arises, is developed in Section 21.

6 A Categorical Interpretation of Spherepop Histories

The representation of objects as event words suggests an immediate connection with category theory, where the primacy of morphisms over objects is a foundational methodological principle [Mac Lane, 1998]. In the categorical framework, processes rather than states are primary; objects emerge as the sources and targets of transformations, and the composition of morphisms encodes the sequential structure of processes. This perspective aligns precisely with the Spherepop principle that identity is grounded in generative history.

Let $\mathcal{C}$ be a category whose objects represent regions or configurations of a system and whose morphisms represent irreversible events. An event

$$E : X \to Y$$

corresponds to a transformation that produces the region Y from the region X. If two events occur sequentially,

$$E_1 : X \to Y, \qquad E_2 : Y \to Z,$$

their categorical composition

$$E_2 \circ E_1 : X \to Z$$

represents the combined historical process producing Z from X. In this formulation, the event word introduced in Section 3 corresponds precisely to a composition of morphisms, and the normal form of an object is the morphism that takes the initial state to the present one.

Branching and merging events, which are prominent features of the worked example, correspond to familiar categorical constructions. A split event

$$E_1 : R \to A \sqcup B$$

resembles a coproduct-like branching in which a single region gives rise to multiple descendants, while a merge event

$$E_3 : (B, C) \to D$$

resembles a pushout-like construction in which two regions produced from a common ancestor are combined into a new one [Mac Lane, 1998, Spivak, 2014]. The precise categorical status of these constructions is examined further in Section 10; for present purposes it suffices to note that the full history of the worked example forms a diagram in $\mathcal{C}$ of the shape illustrated in Figure 2.

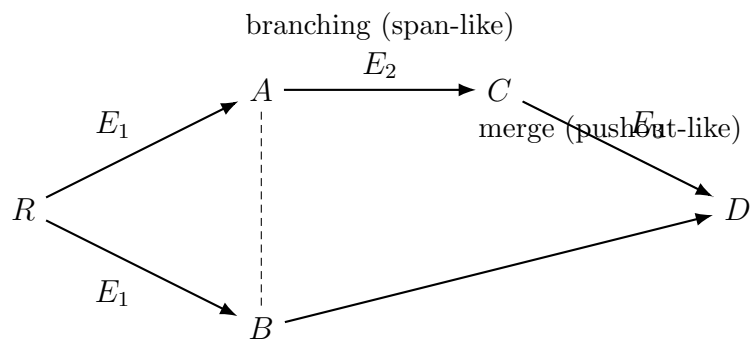

Figure 2: A categorical diagram of the worked example. The split event E_1 produces a span, while the merge event E_3 completes a pushout-like construction. The identity of D is determined by the entire diagram, not merely by the node D in isolation.

From this perspective, the identity of an object is determined by the entire diagram of morphisms that generated it. Two objects are identical precisely when the diagrams representing their histories are identical, up to equality of morphisms. This is a stronger condition than structural isomorphism of the resulting objects; it requires agreement at every stage of the generative process.

The categorical viewpoint also clarifies why Spherepop naturally produces directed acyclic event graphs. Cycles in a morphism-composition diagram would correspond to reversible processes, since a path

$$X \to Y \to \cdots \to X$$

implies the existence of a composite morphism returning from X to itself. In a setting where morphisms represent irreversible transformations, such a cycle would violate

the irreversibility condition. The absence of cycles is therefore not merely a convenient property but a structural consequence of the physical or computational assumption that the events under consideration are irreversible. The relationship between irreversibility and directed acyclicity is developed further in Section 7, and the connection to entropy is examined in Section 5.

Baez and Stay [2010] describe in detail the deep correspondences among physics, topology, logic, and computation through the "Rosetta Stone" of categorical structures; the Spherepop framework sits naturally within the physical and computational columns of that correspondence, where morphisms encode processes and composition encodes sequential execution.

7 Event Graphs and Directed Acyclic Structure

Linear event words provide a correct representation of histories in which each event has at most one predecessor and at most one successor. The processes encountered in practice, however, frequently involve branching and merging: a single region may give rise to multiple descendants, and multiple regions may combine to produce a single new one. In such cases the linear notation, though applicable after a suitable ordering is imposed, obscures the parallel structure of the history. The natural representation is a directed graph.

Definition 7.1 (Event graph)**.** An *event graph* is a directed graph $G = (V, \mathcal{E})$ in which the vertices V represent regions and each directed edge $X \to Y$ represents an irreversible event producing the region Y from the region X.

When events produce multiple descendants or require multiple inputs, the corresponding vertices in G carry multiple outgoing or incoming edges respectively. The complete generative history of a system is encoded in the global structure of G rather than in any single path through it.

Proposition 7.2 (Directed acyclicity)**.** *Every Spherepop event graph is a directed acyclic graph (DAG).*

Proof. Suppose for contradiction that G contains a directed cycle $X \to Y \to \cdots \to X$. Then there exists a composite event morphism that maps X to itself, implying that the transformation can be applied repeatedly to yield a reversible process taking X back to a state indistinguishable from X. Since every Spherepop event is irreversible

by definition, such a composite cannot exist. Consequently no directed cycle can occur in G. $\square$

The directed acyclic property is not merely a convenience; it is the formal analogue of the thermodynamic irreversibility condition formalized in Section 5. Every finite DAG admits a topological ordering, and it is precisely this ordering that will be exploited by the normalization procedure of Section 8 to produce canonical event words from branching histories.

Given an object X appearing as a vertex in G, its identity is determined not by the local properties of the vertex but by the global structure of the subgraph that generated it. Formally, let $\mathrm{Anc}(X)$ denote the set of all vertices from which X is reachable by a directed path in G. The *ancestral subgraph* of X is the induced subgraph

$$G_X = G[\mathrm{Anc}(X) \cup \{X\}]$$

containing all vertices that contributed, directly or indirectly, to the production of X.

Definition 7.3 (Spherepop identity, graphical formulation)**.** Two objects X and Y are Spherepop-identical if and only if their ancestral subgraphs G_X and G_Y are isomorphic as directed graphs with edges labelled by event types.

This definition subsumes the event-word formulation of Section 3 as a special case: when histories are linear (no branching or merging), the ancestral subgraph reduces to a directed path, and graph isomorphism reduces to equality of the corresponding event sequences.

Figure 3 illustrates a more complex event graph containing multiple levels of branching. The terminal object Z is determined by the entire layered DAG structure feeding into it. Different subsets of the graph characterise different terminal objects; the identity criterion requires that the complete ancestral subgraph, not merely the immediately preceding events, be taken into account.

The DAG representation also clarifies the relationship between the event-word notation and the diagrammatic structure. A linear event word corresponds to a directed path through the DAG. Branching corresponds to a vertex with multiple outgoing edges, and merging corresponds to a vertex with multiple incoming edges. The graph therefore provides a complete representation of the causal structure, while the event word provides a linear serialization of that structure. The correspondence between these

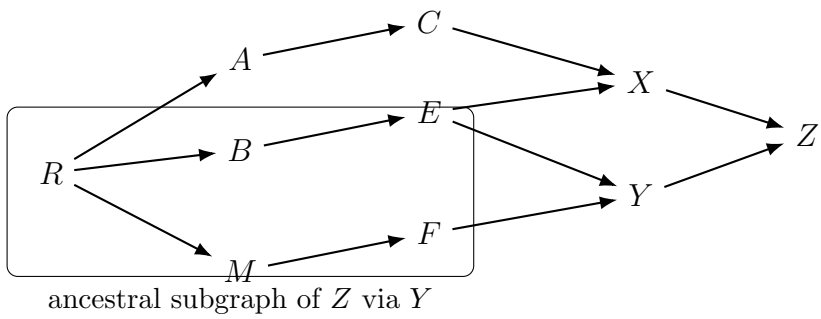

Figure 3: A layered directed acyclic event graph. The identity of the terminal object Z is determined by the entire ancestral subgraph that feeds into it, not merely by its immediate predecessors X and Y.

two representations is the subject of the normalization theory in Section 8, where it is shown that every event graph admits a canonical linear serialization.

8 Normalization and Canonical Representation

The event graph representation captures the full causal structure of a history, but different textual descriptions of the same graph may arise depending on the order in which causally independent events are recorded. If two events act on disjoint regions without either depending on the output of the other, then listing them in either order produces an equivalent description of the same causal structure. This equivalence must be recognized formally if the identity criterion is to be stable under changes of representation.

Definition 8.1 (Event independence)**.** Two events $E_i, E_j \in \mathcal{E}$ are *independent*, written $E_i \parallel E_j$, if neither event requires the output of the other as input; equivalently, there is no directed path between E_i and E_j in the event graph.

Independent events may appear in either order in a linear serialization of the event graph without altering the graph's structure. This motivates the following rewriting rule, which will be made precise in Appendix 31:

$$(E_i, E_j) \longrightarrow (E_j, E_i) \quad \text{whenever } E_i \parallel E_j.$$

Two event words are *causally equivalent* if they are related by a finite sequence of such commutation steps. The normalization procedure selects a canonical representative

from each causal equivalence class.

Definition 8.2 (Spherepop normal form, canonical)**.** The *Spherepop normal form* of an object X is the event word

$$(\text{spherepop}(X))_{\text{nf}}$$

obtained by computing a topological ordering of the event graph G_X and serializing the events in that order, breaking ties consistently according to a fixed deterministic convention on event labels.

Because every finite DAG admits a topological ordering, and because the normalization procedure is deterministic, the normal form is well-defined for every Spherepop object. The canonical identity condition becomes:

$$X \equiv Y \quad \Longleftrightarrow \quad (\text{spherepop}(X))_{\text{nf}} = (\text{spherepop}(Y))_{\text{nf}}.$$

Two objects are identical exactly when their normalized event histories coincide, regardless of the particular serialization that may have been used to describe their histories in the first instance.

Figure 4 illustrates this point concretely. Two syntactically different presentations of the same history—one listing E_i before E_j and the other listing E_j before E_i on two parallel branches—both normalize to the same canonical form, because the two events are independent and the topological ordering convention resolves the tie deterministically.

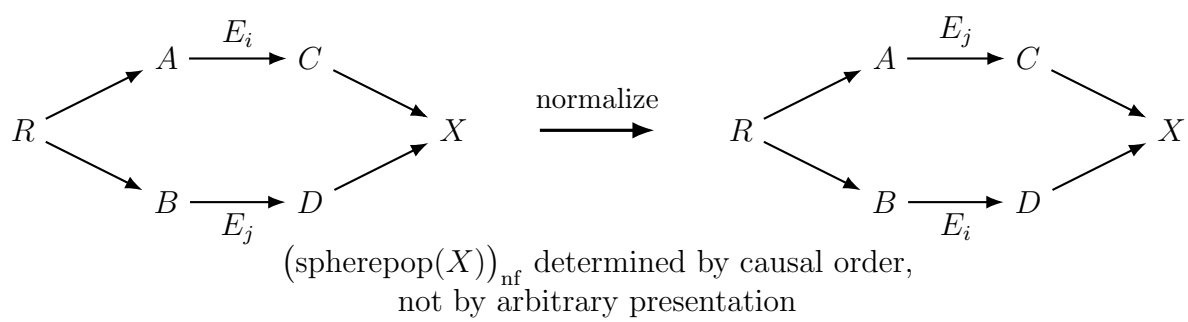

Figure 4: Two syntactically different presentations of independent events E_i and E_j encoding the same causal structure. Normalization collapses both to a single canonical form, since $E_i \parallel E_j$ implies their order may be resolved by any fixed topological convention.

The normalization procedure plays a role in Spherepop analogous to that of canonical forms in other mathematical frameworks: reduced words in group theory, row echelon form in linear algebra, beta-normal forms in the lambda calculus. Each of these devices converts a syntactically diverse family of equivalent expressions into a unique representative that can be compared directly. In Spherepop, the canonical form ensures that identity depends only on causal structure rather than on the order in which an external observer happened to record causally independent events.

9 Functorial Perspectives on Event Histories

The event graph representation and the normalization theory of Section 8 can be cast in a fully categorical language by treating the process of mapping abstract causal structures onto concrete systems as a functor. This perspective, which draws on the categorical foundations developed in Mac Lane [1998] and elaborated in Spivak [2014] and Baez and Stay [2010], provides the most general formulation of Spherepop semantics.

Definition 9.1 (Causal history category)**.** Let $\mathcal{H}$ be a small category whose objects represent intermediate regions or states in a history and whose morphisms represent irreversible events. Because the events are temporally ordered and irreversible, if there exists a morphism $X \to Y$ in $\mathcal{H}$ then there cannot exist a morphism $Y \to X$. Such a category is called a *causal history category*.

In practice, $\mathcal{H}$ may be constructed directly from the event graph G by interpreting vertices as objects and directed edges as generating morphisms, with composition given by directed paths. The temporal ordering that the DAG structure imposes on G corresponds to the "no-back-morphism" condition in $\mathcal{H}$.

Definition 9.2 (Spherepop realization functor)**.** Let $\mathcal{C}$ be a category of regions, configurations, or system states. A *Spherepop realization* of the history $\mathcal{H}$ in $\mathcal{C}$ is a functor

$$F : \mathcal{H} \to \mathcal{C}.$$

The functor F assigns to each node in the abstract causal history a concrete object in $\mathcal{C}$, and to each event morphism the corresponding transformation in the physical or computational system. Functoriality ensures that sequential composition of events in

$\mathcal{H}$ corresponds exactly to sequential composition of transformations in $\mathcal{C}$:

$$F(E_2 \circ E_1) = F(E_2) \circ F(E_1).$$

This coherence condition is the categorical expression of the physical requirement that the combined effect of two sequential events is the composition of their individual effects, without hidden intermediate states.

Terminal objects of $\mathcal{H}$ correspond to the currently existing objects of the system—those that have not been further transformed. If X is such a terminal node, then $F(X)$ represents the concrete present state produced by the entire chain of events that preceded it. The identity of $F(X)$ as a Spherepop object is determined jointly by F and by the position of X in the causal diagram.

Figure 5 illustrates the functorial picture. The abstract causal history category $\mathcal{H}$ on the left carries the relational structure of the history—which events preceded which—without specifying their concrete content. The functor F maps this abstract structure into the category $\mathcal{C}$ of realized regions on the right, where each abstract node becomes a concrete object and each abstract morphism becomes a concrete transformation.

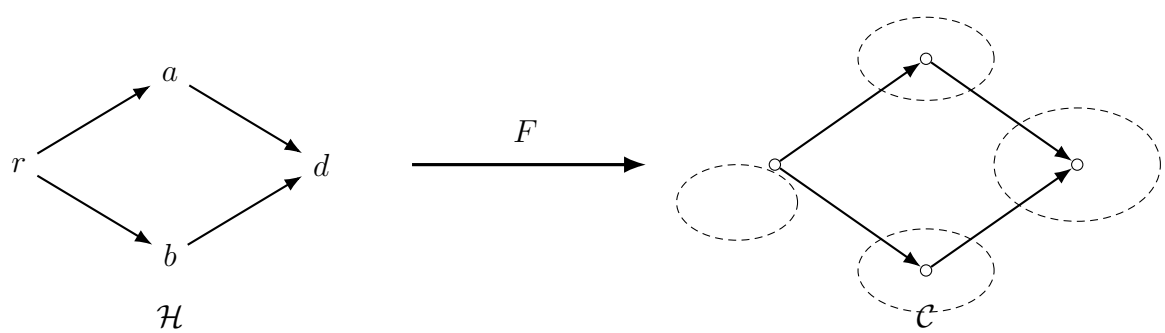

Figure 5: A Spherepop history as a functor $F : \mathcal{H} \to \mathcal{C}$ from an abstract causal index category $\mathcal{H}$ to a category $\mathcal{C}$ of realized regions. The abstract structure on the left encodes causal precedence; the functor translates this into concrete transformations on the right.

The functorial formulation also provides a clean account of when two objects should be considered identical. Suppose

$$F_1 : \mathcal{H}_1 \to \mathcal{C} \quad \text{and} \quad F_2 : \mathcal{H}_2 \to \mathcal{C}$$

are two realizations. The resulting terminal objects coincide as Spherepop objects when

the normalized causal structures $\mathcal{H}_1$ and $\mathcal{H}_2$ are isomorphic as causal categories and the functors agree on the corresponding morphisms under this isomorphism. This condition recovers the normal-form identity criterion of Section 8 in categorical language: it requires agreement not merely on terminal values but on the entire structure of the generating process.

The functorial perspective also suggests how Spherepop should handle abstraction. Natural transformations between realizations—that is, morphisms in the functor category $[\mathcal{H}, \mathcal{C}]$—correspond to systematic reinterpretations of a history that change the concrete objects assigned to nodes while preserving the relational structure. In this way the framework supports a hierarchy of increasingly abstract descriptions of the same underlying causal history.

10 Merge Events as Pushouts

The merge events that appear prominently in Spherepop histories admit a precise categorical characterization as pushout constructions. A pushout in a category $\mathcal{C}$ is the universal solution to the problem of combining two morphisms that share a common source: given morphisms $f : R \to B$ and $g : R \to C$, the pushout is an object D together with morphisms $B \to D$ and $C \to D$ such that the resulting square commutes and D is universal with this property [Mac Lane, 1998].

The connection to Spherepop is direct. In the worked example of Section 2, the regions B and C both originated from the common ancestor R via the split event E_1. The merge event E_3 then combined B and C into the region D. Categorically, this situation is described by the commuting square

$$\begin{tikzcd} R \arrow[r] \arrow[d] & C \arrow[d] \\ B \arrow[r] & D \end{tikzcd}$$

in which D receives consistent morphisms from B and C and is universal with this property. The merge event E_3 corresponds precisely to the completion of this pushout diagram.

The pushout interpretation makes precise the sense in which the identity of D depends on the entire commuting square rather than merely on the pair (B, C) considered in

$$\begin{tikzcd}
 & C \arrow[dr] & \\
R \arrow[ur] \arrow[dr] & \square & D \\
 & B \arrow[ur] &
\end{tikzcd}$$

Figure 6: A merge event represented as a pushout-like square. The common ancestor R gives rise to B and C via the split event; the merge event E_3 completes the square by producing the universal object D.

isolation. Two regions B' and C' could be merged to produce a region D' with the same structural description as D, yet if B' and C' arose from a different common ancestor R', the resulting pushout square would be different and D' would be a different Spherepop object. The pushout construction thus encodes the relational history of the merge event, not merely its output.

This observation connects to the broader theme of the paper. In standard set-theoretic or structural accounts of identity, two sets are equal when they contain the same elements, regardless of how those elements were assembled. The pushout characterization of merge events shows that in Spherepop the corresponding condition is far stronger: the merged object must have arisen from the same causal configuration, with the same common ancestor and the same morphisms connecting it to its predecessors.

The pushout interpretation also interacts with the categorical coherence conditions noted in Section 9. A realization functor $F : \mathcal{H} \to \mathcal{C}$ that preserves pushouts maps each merge event in the abstract history to a genuine pushout in the target category. When $\mathcal{C}$ is a category with all pushouts—for instance, a category of sets or a category of topological spaces—this preservation condition is automatically satisfied. In physical applications, the relevant category may not have all pushouts, and the question of when merge events can be faithfully realized is a substantive one.

11 Entropy Landscapes and Historical Trajectories

A complementary geometric interpretation of Spherepop histories arises from the notion of an entropy landscape. This interpretation is more impressionistic than the category-theoretic treatment of preceding sections, but it connects the formalism to physical intuitions about the role of irreversibility in shaping the space of possible histories.

Let the configuration space of a system be represented as a manifold $\mathcal{M}$ whose points correspond to possible structural states. Define an entropy-like functional

$$S : \mathcal{M} \to \mathbb{R}$$

measuring the informational or thermodynamic complexity of each state. Irreversible events may then be interpreted as transitions that move the system along trajectories in $\mathcal{M}$ while generally increasing the value of S. The history of an object corresponds to a directed path through this landscape, constrained to move in the direction of increasing S.

In this geometric picture, as illustrated in Figure 7, the events E_1, E_2, E_3 correspond to successive steps along a trajectory that climbs the entropy surface. The shape of the landscape determines which transitions are available at each point: a configuration at a local minimum of S that is not a global minimum may be accessible from some prior states but inaccessible from others, and the question of reachability depends on the full path taken up to that point. This path-dependence is precisely the feature that Spherepop captures through its event-word representation: the normal form records not just the endpoint of the trajectory but the entire path.

Critical points of the entropy functional may be interpreted as stable structural configurations or attractor states. The Spherepop history of an object records the sequence of transitions that led to its current position on the landscape, including which saddle points and ridges it passed through on the way. Two objects at the same configuration on the landscape but arriving via different paths are therefore different Spherepop objects, even though a purely spatial description would identify them.

The entropy landscape interpretation also suggests a connection to the theory of gradient flows and Morse theory, where the topology of a functional landscape constrains the structure of trajectories and the classification of critical points. Spherepop histories can in principle be studied using these geometric tools to understand, for instance,

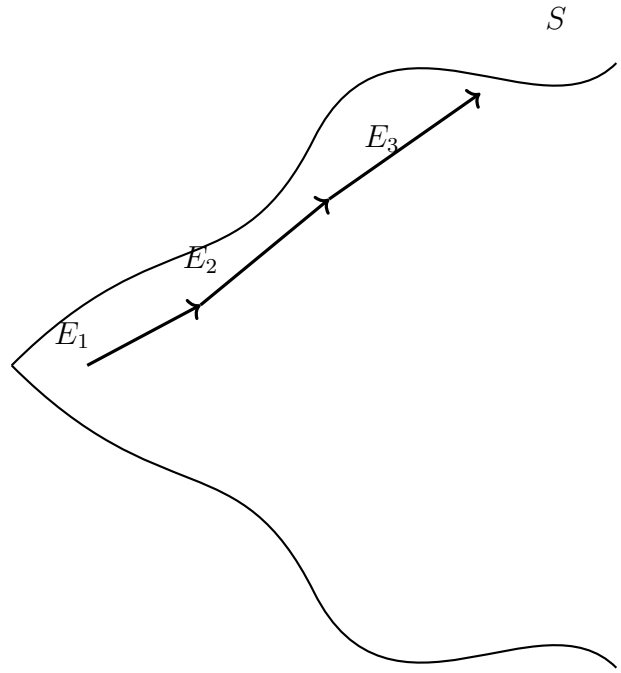

Figure 7: Event history interpreted as a directed trajectory across an entropy landscape. Each event corresponds to a transition along a path that generally moves toward regions of higher entropy. The topology of the landscape constrains which transitions are possible.

which pairs of histories are deformable into one another through the insertion or deletion of null events, or how the topology of the causal DAG interacts with the geometry of the underlying landscape. This direction of generalization lies beyond the scope of the present essay but suggests a rich programme of further investigation.

12 Event Histories as Partial Orders

The directed acyclic structure of Spherepop event graphs implies that each history carries a natural partial order on its events. Making this structure explicit provides a useful bridge between the graph-theoretic and algebraic representations of histories developed in previous sections, and connects the Spherepop framework directly to classical order theory and the concurrency- theoretic tradition of Mazurkiewicz [Mazurkiewicz, 1987].

Let $G = (V, \mathcal{E})$ be an event graph. Define a binary relation $\prec$ on the set of events by

declaring

$$E_i \prec E_j$$

whenever there exists a directed path from E_i to E_j in G. The following proposition is immediate from the DAG property established in Section 7.

Proposition 12.1. *The relation $\prec$ is a strict partial order on the set of events of any Spherepop history.*

Proof. Irreflexivity holds because the event graph is acyclic: no event can appear on a directed path that begins and ends at itself. Asymmetry is immediate from the same acyclicity: if $E_i \prec E_j$ then there is a directed path from E_i to E_j, and the existence of a directed path from E_j to E_i would close a cycle. Transitivity follows by concatenating directed paths: if $E_i \prec E_j$ and $E_j \prec E_k$ then the composed path witnesses $E_i \prec E_k$. □

Each Spherepop history therefore determines a partially ordered set $(\mathcal{E}_X, \prec)$, called the *causal poset* of X. The incomparable elements of this poset are precisely the mutually independent events: two events E_i and E_j satisfy $E_i \parallel E_j$ if and only if neither $E_i \prec E_j$ nor $E_j \prec E_i$.

The event words introduced in Section 3 can now be interpreted with precision as *linear extensions* of the causal poset.

Definition 12.2 (Linear extension)**.** A *linear extension* of $(\mathcal{E}_X, \prec)$ is a total ordering $\leq$ of $\mathcal{E}_X$ such that $E_i \prec E_j$ implies $E_i \leq E_j$.

Every topological ordering of the event graph G_X corresponds to a linear extension of the causal poset, and conversely every linear extension arises from some topological ordering. The collection of all linear extensions therefore coincides precisely with the set of all event words that are causally equivalent in the sense of Section 8: they represent different serializations of the same partial order.

This observation reframes normalization in order-theoretic terms. The commutation rules that identify independent events are precisely the moves that relate different linear extensions of the same poset. The Spherepop normal form selects a canonical linear extension by fixing a deterministic rule for breaking incomparability ties.

The width of the causal poset, which counts the size of the largest antichain, measures the maximal degree of parallelism in the history. Dilworth's theorem [Mac Lane, 1998] states that the minimum number of chains needed to partition the poset equals

the width; in the Spherepop context this means that the most compressed sequential description of a history requires at least as many serialized chains as the width of its causal poset. Wide histories represent highly parallel processes that cannot be faithfully compressed into a single short sequential trace.

The partial-order perspective thus provides the algebraic underpinning for both the normalization theory and the complexity measures developed in the next section, tying these constructions to the classical theory of posets and their linear extensions.

13 Historical Complexity Measures

The entropy measure $S(X)$ introduced in Section 5 captures the cumulative informational weight of a history by summing the contributions of individual events. A complementary family of measures can be derived directly from the structure of the causal poset $(\mathcal{E}_X, \prec)$ introduced in Section 12. These graph-theoretic quantities characterize the structural complexity of a history independently of any specific thermodynamic or informational interpretation.

Definition 13.1 (Historical depth)**.** The *depth* of an object X is the length of the longest directed path in its ancestral event graph:

$$\mathrm{depth}(X) = \max\{\, |p| \mid p \text{ is a directed path in } G_X \,\}.$$

Depth measures the maximal sequential chain of irreversible events separating X from its ultimate ancestor. In a purely linear history depth coincides with the length of the event word, but in branching histories depth may be significantly smaller than the total event count, since many events may proceed in parallel.

Definition 13.2 (Historical width)**.** The *width* of the history of X is the size of the largest antichain in the causal poset:

$$\mathrm{width}(X) = \max\{\, |A| \mid A \subseteq \mathcal{E}(G_X) \text{ is an antichain} \,\}.$$

Width measures the maximal degree of concurrent activity present in the history. A history with large width contains many independent events that could in principle occur simultaneously; a history with width one is entirely sequential.

Definition 13.3 (Branching factor)**.** The *branching factor* of the history of X is the

maximum number of outgoing edges from any vertex in G_X.

Branching events correspond to moments at which the system diversifies into multiple descendant configurations. A high branching factor indicates that individual transformations generate large numbers of descendants, producing rapidly widening event trees.

These three quantities interact in informative ways. By Dilworth's theorem, as noted in Section 12, the minimum number of chains required to cover the causal poset equals the width; consequently $\mathrm{depth}(X) \cdot \mathrm{width}(X)$ provides a lower bound on the total number of events in the history. Conversely, histories in which depth and width are both small are in a precise sense efficiently compressed: they achieve their complexity through neither excessive sequentiality nor excessive concurrency.

Beyond these three measures one may define finer invariants. The *entropy growth rate* of a history is the ratio $S(X)/\mathrm{depth}(X)$, measuring the average thermodynamic cost per sequential step. The *causal density*, defined as the ratio of the number of edges to the number of vertices in G_X, captures how heavily constrained the event graph is by precedence relations. Histories with high causal density are those in which most events depend on most prior events; histories with low causal density are those with sparse causal dependencies and therefore high parallelism.

Together, the complexity measures of this section provide a basic vocabulary for comparing the structural profiles of different generative processes. They complement the entropy functional by describing not the thermodynamic cost of a history but its combinatorial shape—whether it is deep or shallow, narrow or wide, densely or sparsely connected. A full complexity theory of Spherepop histories would develop these measures systematically, establishing bounds on their interactions and identifying which profiles are achievable under given constraints on the event signature.

14 Operational Semantics of Spherepop

The preceding sections have treated Spherepop histories primarily as static structures: event words, partial orders, directed graphs, and categorical diagrams. It is equally important to understand the framework dynamically, as an operational system in which configurations evolve through discrete transitions. This section introduces a small-step operational semantics for Spherepop that makes the execution model explicit and

connects the historical identity criterion to the standard vocabulary of process calculi and labeled transition systems [Abramsky and Jung, 1994].

Let Σ denote the set of possible system configurations. A *labeled transition system* (LTS) for Spherepop consists of a set Σ of configurations, a set $\mathcal{E}$ of event labels, and a transition relation

$$\xrightarrow{E} \;\subseteq\; \Sigma \times \mathcal{E} \times \Sigma,$$

written $X \xrightarrow{E} Y$ when the event E transforms configuration X into configuration Y. The irreversibility assumption requires that the relation contain no pair of transitions $X \xrightarrow{E} Y$ and $Y \xrightarrow{E'} X$ for any events E and E'.

A Spherepop history arises as an execution trace of this LTS. Starting from an initial configuration X_0, a sequence of enabled transitions

$$X_0 \xrightarrow{E_1} X_1 \xrightarrow{E_2} X_2 \xrightarrow{E_3} \cdots \xrightarrow{E_k} X_k$$

generates the object X_k. The normal form of X_k is the trace of event labels produced by this execution:

$$(\text{spherepop}(X_k)) = (E_1, E_2, \ldots, E_k).$$

This connection between the static normal form and the dynamic execution trace is the operational content of the Spherepop identity criterion: two objects are identical precisely when no execution producing one can be distinguished from any execution producing the other by examining their event-label sequences.

Branching configurations arise when a single configuration X admits multiple simultaneously enabled transitions acting on disjoint parts of the system. In such cases, rather than choosing an order of execution, the transitions may proceed concurrently. The resulting structure is no longer a linear trace but the directed acyclic event graph described in Section 7, whose vertices are intermediate configurations and whose edges are individual transitions. The normalization of Section 8 selects a canonical serialization of this concurrent execution, corresponding to a particular scheduling order for the independent transitions.

This operational picture also clarifies the role of the empty event word ε introduced in Section 3. In the LTS formulation, ε corresponds to the identity trace: an object in the initial configuration X_0 that has undergone no transitions. Such an object is

defined entirely by its starting state, with no generative history to record.

The relationship between the Spherepop operational semantics and standard bisimulation equivalences is instructive. In the theory of process calculi, two processes are bisimilar when each can simulate the other step by step [Abramsky and Jung, 1994]. The Spherepop identity criterion is strictly stronger than bisimilarity: it requires not merely that two objects respond to future transitions in the same way, but that they arrived at their present state via identical histories. Bisimulation looks forward; Spherepop identity looks backward. This contrast mirrors the broader distinction between structural and historical identity discussed in Section 16.

15 The Historical Reconstruction Problem

The operational semantics of Section 14 raises a natural inverse problem. Given an object X in its present configuration, is it possible to recover the history that produced it? In classical Hamiltonian mechanics, reversibility ensures that the present state uniquely determines all past states. Spherepop, by contrast, is grounded in irreversibility. The question of historical reconstruction therefore has a fundamentally different character.

Definition 15.1 (Historical reconstruction problem)**.** Given a present object X with configuration C, the *historical reconstruction problem* asks for the set

$$\mathcal{R}(X) = \{\, H \mid H \text{ is an event history producing a configuration isomorphic to } C \,\}.$$

In general $\mathcal{R}(X)$ is not a singleton. Multiple distinct histories may converge to structurally indistinguishable configurations, as the worked example of Section 2 illustrates: the objects D and D' may be structurally identical even though their generating histories are distinct. The reconstruction problem therefore typically admits infinitely many solutions, corresponding to the infinitely many event sequences that could in principle have generated the observed configuration.

This indeterminacy is not a defect of the formalism but a reflection of an informational fact about irreversible processes. Each irreversible event dissipates information about the prior configuration, so that the prior state is in general not recoverable from the posterior state alone. The Spherepop normal form resolves this ambiguity by treating the history as a primitive datum: the identity of X is not the configuration C but the

specific history H that produced it.

From the perspective of causal inference, the reconstruction problem corresponds to asking which causal structures are consistent with the observed data. In many applications, partial information about the history may be available from side channels—for instance, timestamps, audit logs, or provenance metadata—and the problem reduces to disambiguating among histories consistent with this partial information. The Spherepop framework provides a formal language in which such disambiguation can be stated: one seeks a history H satisfying both the structural constraint $H \in \mathcal{R}(X)$ and any additional constraints imposed by the available side-channel information.

The reconstruction problem also has thermodynamic significance. The number of distinct histories in $\mathcal{R}(X)$ may be interpreted as an entropy of the set of possible pasts compatible with the present observation. A large set $\mathcal{R}(X)$ indicates that the present state is informationally impoverished with respect to the past: many different processes could have generated it. A singleton $\mathcal{R}(X)$ would indicate a maximally informative present state, one from which the unique generating history can be fully recovered—but in an irreversible system this is the exceptional rather than the typical case.

By foregrounding this asymmetry, the Spherepop framework aligns with a broadly held principle in the philosophy of time: the past is determined and the future is open, but the past is often informationally inaccessible from the present. The normal form preserves the information that would otherwise be lost through irreversibility, precisely by treating it as constitutive of identity rather than as a contingent annotation.

16 Identity as Projected History

The preceding sections have established two complementary formal structures. Event words and normal forms give every object a canonical symbolic encoding; identity is determined by the equality of those encodings. At a deeper level, event words arise as the projection of trajectories through a space of possibility-history pairs, and Section 4 showed that the history projection Π_{hist} maps each trajectory to its accumulated commitment record. The present section unifies these two levels by reformulating identity directly in terms of trajectories and their projection, making precise the sense in which identity is not a primitive property of objects but an equivalence class defined by an irreversible process.

16.1 The Trajectory Category

Let $\mathcal{T}$ denote the category of admissible trajectories. Objects of $\mathcal{T}$ are Spherepop states (Ω_t, H_t), and morphisms are irreversible extensions: a morphism from (Ω_t, H_t) to $(\Omega_{t'}, H_{t'})$ exists if and only if $\Omega_{t'} \subseteq \Omega_t$, $H_{t'} = H_t \cdot w$ for some non-empty w, and the extension satisfies the axioms of Section 4. Composition is concatenation of extensions.

The trajectory category has no non-trivial isomorphisms: since Ω can only shrink and H can only grow, no morphism from $(\Omega_{t'}, H_{t'})$ back to (Ω_t, H_t) can exist unless the two states are identical. This is the categorical expression of irreversibility at the dynamical level.

The history projection

$$\Pi_{\text{hist}} : \mathcal{T} \longrightarrow \mathbb{P}(\Omega)$$

is a functor that maps each state (Ω_t, H_t) to the event word H_t and maps each morphism (extension) to the corresponding prefix extension in $\mathbb{P}(\Omega)$. Functoriality is immediate: composing two extensions in $\mathcal{T}$ corresponds to concatenating their event words, which is exactly composition in $\mathbb{P}(\Omega)$.

16.2 Projected Identity

The Spherepop identity criterion may now be stated in its most general form.

Definition 16.1 (Projected identity)**.** Two terminal states (Ω_k, H_k) and (Ω'_k, H'_k), arising from trajectories τ and τ' in $\mathcal{T}$, are *Spherepop-identical* if and only if their projected histories coincide:

$$\Pi_{\text{hist}}(\tau) \;=\; \Pi_{\text{hist}}(\tau').$$

Identity is thus defined not directly on trajectories but on their images under Π_{hist}. Distinct trajectories—ones that trace different paths through option space, eliminating possibilities in different intermediate orders—may still produce identical objects if the ordered sequence of committed events is the same. The object is the equivalence class; the trajectory is one representative of it.

Definition 16.2 (Trajectory equivalence)**.** For trajectories $\tau, \tau' \in \mathcal{T}$, define

$$\tau \sim \tau' \iff \Pi_{\text{hist}}(\tau) = \Pi_{\text{hist}}(\tau').$$

The objects of the Spherepop ontology are equivalence classes $[\tau]$ under this relation. Each class consists of all trajectories that produce the same ordered boundary trace, and the event word is the minimal complete invariant of that class: it preserves all distinctions relevant to identity and loses nothing that matters.

16.3 Compatibility with Normal Forms

This reformulation is fully compatible with the earlier event-word formalism. Let G_X be the ancestral event graph associated with a trajectory τ, and let $(\text{spherepop}(X))_{\text{nf}}$ be its canonical linearisation obtained by the normalization procedure of Section 8. The projection satisfies

$$\Pi_{\text{hist}}(\tau) \;=\; (\text{spherepop}(X))_{\text{nf}},$$

so the previously defined identity condition

$$X \equiv Y \iff (\text{spherepop}(X))_{\text{nf}} = (\text{spherepop}(Y))_{\text{nf}}$$

is recovered as the special case of projected identity in which histories have already been normalised to canonical form. The normal form is the canonical representative of the equivalence class $[\tau]$ in $\mathbb{P}(\Omega)$.

16.4 Non-Injectivity and Loss of Preimage Information

The projection Π_{hist} is not injective on objects of $\mathcal{T}$.

Proposition 16.3 (Non-injectivity of history projection)**.** *There exist distinct trajectories $\tau \neq \tau'$ in $\mathcal{T}$ such that $\Pi_{\text{hist}}(\tau) = \Pi_{\text{hist}}(\tau')$.*

Proof. Let $\Omega_0 = \{a, b, c\}$ and consider two trajectories that both commit event a first and event b second. In the first trajectory, the option space evolves as $\{a, b, c\} \to \{b, c\} \to \{c\}$. In the second, a **Bind** constraint removes c before the first commitment, so the option space evolves as $\{a, b, c\} \to \{a, b\} \to \{b\} \to \varnothing$ with the binding step making c unavailable. Both trajectories project to the history (a, b) under Π_{hist}, yet they trace different paths through $\mathcal{T}$ because the intermediate option spaces differ. $\square$

This non-injectivity formalises a central feature of the framework. Identity preserves the order of actualisations but not the full geometry of the possibility space from which that order emerged. The preimage of an event word under Π_{hist} consists of all possible

ways in which that sequence of exclusions could have been realised: different initial option spaces, different structural constraints, different available alternatives that were passed over. None of this information is retained in the identity of the resulting object.

This is precisely what it means for the history to be the irreversible trace. The information lost in passing from trajectory to history is the information about alternatives: the options that were available but not taken, the paths through possibility space that were open but not followed. Once the trajectory has terminated in an event word, those alternatives cannot be recovered from the word alone. The historical reconstruction problem of Section 15 is the formal expression of this impossibility at the level of states; the non-injectivity of Π_{hist} is its expression at the level of trajectories.

16.5 Relationship to Structural Identity

The reformulation as projected identity also clarifies the relationship to structural equivalence, which was discussed informally in the earlier version of this section. Structural identity corresponds to a further projection: from the event word to the present observable state via the Collapse functor. The chain

$$\mathcal{T} \xrightarrow{\Pi_{\text{hist}}} \mathbb{P}(\Omega) \xrightarrow{\textbf{Collapse}} \mathcal{S}$$

gives three levels of identity, each coarser than the last. At the level of $\mathcal{T}$, identity requires full trajectory equivalence. At the level of $\mathbb{P}(\Omega)$, identity requires only equal event words. At the level of $\mathcal{S}$, identity requires only equal observable states.

These correspond precisely to the stratified equivalence relations of Section 18: Spherepop identity sits between trajectory equivalence and structural isomorphism, retaining the causal order of events while discarding the full dynamics of the option space.

The distinction illustrated in Figure 8 now has an explicit categorical interpretation. The two histories H_1 and H_2 correspond to distinct equivalence classes in $\mathbb{P}(\Omega)$ under Π_{hist}, and they remain distinct under any identity criterion that does not further apply **Collapse**. Structural identity is the criterion that applies both projections; Spherepop identity is the criterion that applies only the first.

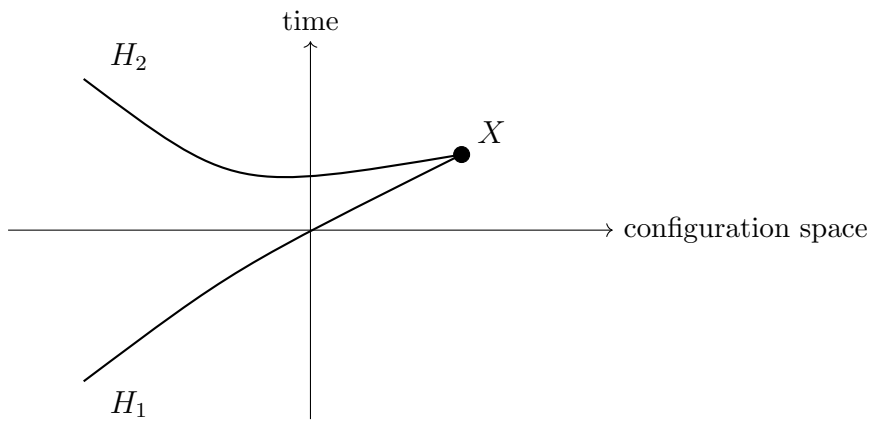

Figure 8: Two histories H_1 and H_2 converging to the same terminal configuration. In the trajectory-based formulation, these represent distinct equivalence classes in $\mathcal{T}$ that project to distinct objects in $\mathbb{P}(\Omega)$ (since the event words differ) even though both project to the same object in $\mathcal{S}$ under **Collapse**. Structural identity conflates them; Spherepop identity does not.

16.6 Identity as a Quotient of Irreversibility

The formulation of identity as projected history makes explicit why irreversibility is not merely a useful property of the framework but a constitutive one. Because $\mathcal{T}$ has no non-trivial isomorphisms, the projection Π_{hist} is a quotient map onto an equally irreversible category. The equivalence relation $\sim$ on $\mathcal{T}$ that defines identity is induced by this projection, and its structure is determined entirely by which information Π_{hist} preserves and which it discards.

What is preserved: the set of committed events and the order in which they occurred. What is discarded: the intermediate structure of the option spaces, the paths through possibility that were available but not taken, and any information about the geometry of the contraction that is not encoded in the ordered sequence of its outputs.

Identity is therefore a quotient of an irreversible process. It retains the temporal ordering of commitments—because that ordering is the substance of the event word—but forgets the full space of alternatives that preceded each commitment. This quotient is not arbitrary or chosen for convenience; it is forced by the structure of Π_{hist} and by the irreversibility of the dynamics in $\mathcal{T}$ that the projection collapses.

17 Irreversibility and the Necessity of Historical Identity

The preceding sections have developed historical identity as a representational choice—a richer framework than extensional identity that preserves the causal record which extensional accounts discard. But the relationship between the two is stronger than a mere difference in descriptive power. Under conditions of genuine irreversibility, extensional identity fails as a faithful criterion: it identifies objects that are, in any causally or computationally significant sense, distinct. This section makes that failure precise.

Definition 17.1 (Extensional equivalence)**.** Two histories $H, H' \in \mathcal{E}^*$ are *extensionally equivalent* if they produce the same observable outcome under the Collapse functor:

$$H \sim_{\text{ext}} H' \quad \Longleftrightarrow \quad \text{Collapse}(H) = \text{Collapse}(H').$$

Definition 17.2 (Intensional identity)**.** Two histories are *intensionally identical* if they are literally equal as ordered sequences, or equivalently, if their Spherepop normal forms coincide:

$$H =_{\text{int}} H' \quad \Longleftrightarrow \quad (\text{spherepop}(H))_{\text{nf}} = (\text{spherepop}(H'))_{\text{nf}}.$$

The inclusion $=_{\text{int}} \subseteq \sim_{\text{ext}}$ is immediate: intensionally identical histories produce the same observable outcome, since Collapse is a functor. The converse fails in general, and the failure is not accidental.

Theorem 17.3 (Failure of Extensional Identity under Irreversibility)**.** *If the event signature contains elements a, b such that $T_a \circ T_b \neq T_b \circ T_a$ for some root state s_0, then extensional equivalence does not define a faithful identity relation on histories: there exist histories $H \neq H'$ with* $\text{Collapse}(H) = \text{Collapse}(H')$ *and histories $H_1 \neq H_2$ with* $\text{Collapse}(H_1) \neq \text{Collapse}(H_2)$*. In particular, no extensional criterion can recover the full intensional structure of the history category.*

Proof. For the first claim, choose events a and b that happen to commute at a particular point: $T_a \circ T_b(s_0) = T_b \circ T_a(s_0)$. The histories $H = (a, b)$ and $H' = (b, a)$ are distinct as ordered sequences, hence $H \neq H'$ under intensional identity, yet $\text{Collapse}(H) = \text{Collapse}(H')$, so they are extensionally equivalent. Any identity criterion that relies

solely on observable output collapses a genuine distinction.

For the second claim, choose a, b such that $T_a \circ T_b(s_0) \neq T_b \circ T_a(s_0)$, which is possible by hypothesis. Then $H_1 = (a, b)$ and $H_2 = (b, a)$ are extensionally distinct. Since they are also intensionally distinct, extensional equivalence is both coarser and finer than intensional identity in different parts of the history space: it conflates some distinct histories while distinguishing others. No quotient of the extensional equivalence classes faithfully recovers the intensional structure. □

Corollary 17.4 (Necessity of Historical Identity)**.** *In any irreversible system in which commitment order is causally significant, the only identity criterion that is simultaneously faithful and compositionally coherent is intensional historical identity.*

The philosophical weight of this result is substantial. It transforms the Spherepop framework from a representational preference into a logical necessity. The argument is not that historical identity is *useful* for recording provenance—which it clearly is—but that in a world of genuine irreversibility, extensional identity is the wrong notion of identity, and historical identity is the only consistent alternative.

This is precisely the position that the Spherepop framework has been developing from the opening sections. Two objects whose present configurations agree but whose histories differ are not merely described differently by the two criteria; they are genuinely different objects, and any framework that fails to distinguish them has changed its subject. The worked example of Section 2, the DAG structure of Section 7, the operational semantics of Section 14, and the reconstruction analysis of Section 15 all instantiate the same underlying point that Theorem 17.3 makes precise: history cannot be recovered from state, and a theory that identifies by state has lost what matters.

18 Equivalence Relations on Histories

The Spherepop identity criterion identifies two objects precisely when their normalized event histories coincide. This is the strictest reasonable notion of identity within the framework. However, many applications require coarser comparisons—situations in which two objects are considered equivalent even though their detailed histories differ. This section introduces a stratified family of equivalence relations that captures the range of positions between strict historical identity and pure structural indistinguishability.

The finest relation is Spherepop identity itself, which has been the focus of the preceding sections. Two objects are Spherepop-identical when their normal forms are equal as event words:

$$X \equiv_{\mathrm{sp}} Y \quad \Longleftrightarrow \quad (\mathrm{spherepop}(X))_{\mathrm{nf}} = (\mathrm{spherepop}(Y))_{\mathrm{nf}}.$$

A strictly coarser relation is *causal isomorphism*, which identifies objects when their ancestral event graphs are isomorphic as labeled directed graphs but does not require that the vertex sets be literally equal:

$$X \equiv_{\mathrm{ci}} Y \quad \Longleftrightarrow \quad G_X \cong G_Y.$$

Under causal isomorphism two objects are equivalent when they arose from histories with the same causal shape and the same event types at each node, even if the specific identifiers of intermediate regions differ. This relation is useful in contexts where the labels of intermediate states are considered implementation details rather than constitutive features of identity.

Coarser still is *trace equivalence*, which identifies objects when their sets of possible executions—their sets of linear extensions of the causal poset—coincide:

$$X \equiv_{\mathrm{tr}} Y \quad \Longleftrightarrow \quad \mathrm{Lin}(G_X) = \mathrm{Lin}(G_Y),$$

where $\mathrm{Lin}(G)$ denotes the set of all linear extensions of the event graph G. Trace equivalence captures the idea that two objects are indistinguishable with respect to every possible sequential observation of their histories, while still retaining causal information beyond what is present in the final configuration.

The coarsest relation in this family is *structural isomorphism*, which identifies objects when their present configurations are isomorphic without any reference to history:

$$X \equiv_{\mathrm{struct}} Y \quad \Longleftrightarrow \quad C_X \cong C_Y,$$

where C_X denotes the present configuration of X. This is the set- theoretic notion of extensional identity discussed in Section 16, which discards all historical information.

These four relations are ordered by inclusion:

$$\equiv_{\text{sp}} \subseteq \equiv_{\text{ci}} \subseteq \equiv_{\text{tr}} \subseteq \equiv_{\text{struct}} .$$

Each successive relation is strictly coarser in general, though specific examples may make some pairs coincide. For instance, in a system with no branching, all linear event words with the same length over the same alphabet are already trace-equivalent, so $\equiv_{\text{sp}}$ and $\equiv_{\text{tr}}$ coincide.

The choice of which equivalence relation to employ depends on the application. For auditing and accountability, full Spherepop identity is appropriate. For type-theoretic reasoning about program behavior, causal isomorphism may suffice. For behavioral equivalences in process calculi, trace equivalence connects naturally to the Mazurkiewicz tradition [Mazurkiewicz, 1987]. For extensional mathematics, structural isomorphism suffices. The Spherepop framework provides the vocabulary to make these distinctions explicit and to reason about what is preserved or lost under each.

19 Spherepop as a Free Traced Monoidal Construction

The compositional structure of Spherepop histories places the framework naturally within the theory of categorical process theory. The primitive operations of the formalism—the sequential composition of events, the independent parallel occurrence of events, and the merging and splitting constructions—correspond to the standard generators of a symmetric monoidal category. When the additional structure of internal closure is taken into account, the framework extends to a traced monoidal category. This section makes these connections explicit and uses them to clarify the universal properties of the Spherepop construction.

Let Σ be a generating signature consisting of primitive event types. Each event type in Σ is specified by the interface it consumes and the interface it produces, where an interface is a collection of regions that the event acts upon. From Σ one constructs the free symmetric monoidal category $\mathcal{F}(\Sigma)$ in which morphisms correspond to event histories assembled from primitive events by sequential composition and monoidal product.

Sequential composition of events corresponds to categorical composition: if $E_1 : A \to$

B and $E_2 : B \to C$ are event types, then their sequential occurrence produces the composite morphism $E_2 \circ E_1 : A \to C$. The monoidal product $E_1 \otimes E_2 : A_1 \otimes A_2 \to B_1 \otimes B_2$ corresponds to the independent parallel occurrence of E_1 and E_2 acting on disjoint interfaces. An event word

$$(E_{i_1}, E_{i_2}, \ldots, E_{i_k})$$

therefore corresponds to the composite morphism

$$E_{i_k} \circ \cdots \circ E_{i_2} \circ E_{i_1}$$

in $\mathcal{F}(\Sigma)$. Branching and merging events, when represented as string diagrams in $\mathcal{F}(\Sigma)$, appear as boxes with multiple output wires and multiple input wires respectively, following the standard graphical calculus for monoidal categories [Baez and Stay, 2010].

The commutation rules of Section 8, which identify event words that differ only by the ordering of independent events, correspond precisely to the symmetry isomorphisms of the monoidal structure. In the free symmetric monoidal category, the symmetry natural transformation

$$\sigma_{A,B} : A \otimes B \xrightarrow{\sim} B \otimes A$$

witnesses the equivalence between the two orderings of independent events. The Spherepop normal form selects a canonical representative from each such equivalence class, mirroring the canonical form of morphisms in the free symmetric monoidal category modulo the coherence axioms.

Many Spherepop histories, however, involve intermediate regions that are created in one part of the history and subsequently consumed by a later merge event. Such patterns give rise to internal wires in the string diagram representation: wires that connect an output of one box to an input of a later box without being visible at the external boundary of the entire diagram. When such internal wires are systematically hidden, the appropriate categorical structure is a *traced monoidal category* [Baez and Stay, 2010].

In a traced symmetric monoidal category, the trace operation

$$\mathrm{Tr}^U_{A,B} : \mathrm{Hom}(A \otimes U,\, B \otimes U) \to \mathrm{Hom}(A, B)$$

formalizes the process of routing an intermediate wire of type U from one stage of a diagram back into a later stage while hiding it from the external interface. The resulting morphism from A to B retains only the observable boundary behavior, abstracting over the internal routing.

Spherepop merge events, and more generally any situation in which intermediate regions are created and subsequently absorbed, can be interpreted as trace operations applied to the underlying string diagram. A history in which a region U is produced by one event and consumed by a later merge event contributes a traced component to the diagram, even though U may not appear in the final configuration.

The Spherepop framework may therefore be interpreted as the free traced symmetric monoidal category $\mathcal{F}_{\mathrm{tr}}(\Sigma)$ generated by the event signature Σ. The universal property of this construction states that any assignment of concrete morphisms to the primitive event types, respecting the traced monoidal structure of the target category, extends uniquely to a traced monoidal functor from $\mathcal{F}_{\mathrm{tr}}(\Sigma)$. Every Spherepop realization functor of Section 9 is an instance of this universal property.

What distinguishes Spherepop from an abstract categorical formalism is its ontological interpretation. In an ordinary traced monoidal category, morphisms represent processes and objects represent interfaces. In Spherepop, the morphisms are not merely descriptions of processes but constitute the identities of the objects they produce. A Spherepop object is an output boundary *together with* its generating morphism class in the free traced monoidal category. Two output boundaries are the same Spherepop object if and only if they are generated by the same morphism.

This identification of objects with their generating morphisms is the categorical formulation of the historical identity criterion developed throughout the paper. It situates the Spherepop ontology within the established framework of categorical process theory while emphasizing the philosophical commitment that distinguishes it from purely operational formulations of that theory.

20 Provenance Graphs and Distributed Event Logs

The Spherepop framework bears a close relationship to provenance tracking systems used in distributed computation and data management. In such systems the central problem is to record not merely the current state of a dataset but the sequence of

operations that produced it. The resulting provenance graph serves as a historical record that allows users to verify, reproduce, and audit the transformations that generated a given artifact. This section examines the connection between Spherepop and provenance systems, distributed event logs, and blockchain-style ledgers, arguing that Spherepop provides a mathematical abstraction of patterns that already appear in practical systems.

A provenance graph typically consists of three categories of entities: data objects, transformation processes, and dependency relations [Spivak, 2014]. Processes consume input objects and produce output objects, and the directed edges of the graph record these causal dependencies. The structure that results is a directed acyclic graph representing the lineage of every object in the system. Spherepop histories may be interpreted as a formal abstraction of exactly this structure: regions correspond to data objects, events correspond to transformation processes, and the ancestral subgraph G_X defined in Section 7 is precisely the provenance graph of the object X.

This interpretation makes precise a practical motivation for the Spherepop identity criterion. In many data-management contexts it is insufficient to know that two datasets are structurally identical; one must also know whether they were produced by the same sequence of transformations. Two files that contain identical data may nevertheless be considered distinct artifacts if they were produced by different computational workflows, because their different workflows may carry different validity guarantees, different institutional authorizations, or different reliability expectations. The Spherepop criterion captures this distinction exactly: two objects are identical only when their entire generative histories coincide.

Distributed event logs provide a closely related example. In modern distributed systems each component records a sequence of events representing local state transitions, and the global behavior of the system is reconstructed by combining these logs into a partial order of events, typically using logical clock mechanisms or causal message relationships. The structure that results is again a directed acyclic event graph, precisely of the form studied in Section 7. The causal poset of Section 12 formalizes the partial order that distributed systems practitioners construct from their logs, and the normalization procedure of Section 8 corresponds to the operation of computing a canonical linearization of concurrent events for purposes of comparison or replay.

Blockchain systems provide a particularly vivid illustration of the same principle. In a

blockchain ledger the identity of a transaction is inseparable from its position within the chain of prior transactions. The ledger records an immutable append-only sequence of state transitions, and the validity of each entry depends on the entire chain of entries that preceded it. The identity of any ledger state is therefore constituted by the complete history of appended events, not by the final state alone. This is the Spherepop identity criterion applied to a specific computational substrate.

These examples suggest that Spherepop is not merely a theoretical construct but a mathematical articulation of patterns that arise independently in engineering practice. The framework provides a language for expressing precisely what these systems mean when they record a history, for reasoning about when two recorded histories represent the same event structure, and for studying the invariants that are preserved under different serialization and compression strategies. The normalization theory of Section 8 and the rewriting system of Appendix 31 apply directly: the question of whether two distributed logs represent the same global computation is precisely the question of whether their corresponding event words normalize to the same Spherepop normal form.

21 Operators as Generators of Event Words

The state-pair dynamics introduced in Section 4 and the entropy structure derived in Section 5 establish that event words arise from a deeper process of successive exclusion. The present section enriches the event alphabet $\mathcal{E}$ by giving it internal structure: rather than treating events as undifferentiated elements of a set, we show that every event can be classified as one of four fundamental generator types, each corresponding to a distinct mode of commitment. This classification does not change the formal machinery of event words, DAGs, or normalization; it adds a typing layer that distinguishes *how* an exclusion occurred from the fact that it occurred.

21.1 Four Modes of Exclusion

The following operators are derived from the axioms of Section 4, not introduced as independent primitives. Each is a specific pattern of state-pair transition.

Pop: Irreversible Commitment. The most fundamental generator. For an event $x \in \Omega_t$, a **Pop**-transition removes x from the option space and appends it to the history

as a deliberate, chosen commitment:

$$\mathbf{Pop}_x(\Omega_t, H_t) \;=\; \big(\Omega_t \setminus \{x\},\; H_t \cdot x\big).$$

This is the generator corresponding to an ordinary event $E \in \mathcal{E}$ in the event-word formalism. Every event in a Spherepop history is, at minimum, a **Pop** transition. There is no $\mathbf{Pop}^{-1}$, because by Axiom 4.2 no operation can restore an excluded event.

Bind: Contextual Restriction. A structural narrowing of the option space that leaves the history unchanged. For an admissibility predicate $f : \mathcal{E} \to \{0, 1\}$:

$$\mathbf{Bind}_f(\Omega_t, H_t) \;=\; \big(\{x \in \Omega_t \mid f(x) = 1\},\; H_t\big).$$

Bind does not append to the event word. It models the narrowing of admissible events by structural constraint or environmental condition before any deliberate selection occurs. The entropy weight w of Section 5 registers the reduction in $|\Omega_t|$, but the history is unchanged, so the event word carries no direct record of the constraint. This is precisely the distinction between *structural impossibility* and *chosen commitment*, which matters for the responsibility weights introduced below.

Refuse: Principled Exclusion. Geometrically identical to **Pop** in its effect on the state pair, **Refuse** differs in the annotation attached to the appended event:

$$\mathbf{Refuse}_x(\Omega_t, H_t) \;=\; \big(\Omega_t \setminus \{x\},\; H_t \cdot (x, \rho_x)\big),$$

where $\rho_x \neq 0$ is the responsibility weight recording that this exclusion was an act of principled negation rather than instrumental selection. The event word records the exclusion; the annotation records its character. In the typed event alphabet, a **Refuse** event is an event of type Refused rather than type Committed, which means that the equivalence relations of Section 18 can be refined to respect this typing: two histories may be trace-equivalent with respect to the raw event sequence but intensionally distinct with respect to the pattern of refusals they contain.

Collapse: Reconstruction from History. Not a transition but a projection. Given a fixed root state s_0 and a family of transformations $\{T_x\}_{x \in \mathcal{E}}$, the **Collapse** operator

is the map that realizes the observable state from the committed history:

$$\mathbf{Collapse}(H_t) \;=\; T_{x_n} \circ \cdots \circ T_{x_1}(s_0) \quad \text{for } H_t = (x_1, \ldots, x_n).$$

This is the same functor introduced in Section 9 under the name of the realization functor $F : \mathcal{H} \to \mathcal{C}$. Naming it **Collapse** emphasises its role in the state-pair framework: it is the operation that recovers an observable shadow from the history. The observable state is never stored as a primary object; it is always derived on demand from the commitment record. This inversion of the usual architecture—history is primary, state is derived—is the computational expression of the ontological commitment established in Section 4.

21.2 Typing the Event Alphabet

With these four operator types identified, the event alphabet $\mathcal{E}$ can be given a type-theoretic structure. Define:

$$\mathcal{E} \;=\; \mathcal{E}_{\mathbf{Pop}} \;\cup\; \mathcal{E}_{\mathbf{Refuse}} \;\cup\; \mathcal{E}_{\mathbf{Bind}},$$

where the three sets are not required to be disjoint at the level of underlying events but carry distinct semantic annotations. A **Collapse** event is not an element of $\mathcal{E}$ at all; it is a derived operation on the history. Event words are then typed sequences of elements from this partitioned alphabet, and the normalization procedure of Section 8 can be refined to respect the typing: two events are independent (and hence their order may be commuted in normalization) only if they are independent both as elements of $\mathcal{E}$ in the causal sense and as bearers of the same type in the semantic sense.

This typing enriches the equivalence relations of Section 18 without disrupting them. The raw causal equivalences—Spherepop identity, causal isomorphism, trace equivalence, structural isomorphism—remain defined as before. But a new dimension of equivalence is now available: responsibility-preserving equivalence, which identifies two histories only when they agree not only on the causal order of events but on the type (**Pop** versus **Refuse**) of each event. This is the finest equivalence in the full lattice, sitting strictly above Spherepop identity in discriminatory power.

21.3 Responsibility as a Weighting on Contractions

The entropy weight $w(E)$ introduced in Section 5 measures the magnitude of the possibility-space contraction associated with each event. The responsibility weight $\rho(E)$ introduced in that section's final subsection provides a complementary measure. We can now be precise about what ρ measures.

Definition 21.1 (Responsibility weight)**.** The *responsibility weight* of an event E in a history is defined by

$$\rho(E) \;=\; \begin{cases} 0 & \text{if } E \in \mathcal{E}_{\mathbf{Bind}}, \\ \lambda & \text{if } E \in \mathcal{E}_{\mathbf{Pop}}, \\ \mu & \text{if } E \in \mathcal{E}_{\mathbf{Refuse}}, \end{cases}$$

for non-negative constants λ and μ with $0 < \lambda \leq \mu$. The total responsibility of an object is

$$\mathcal{R}(X) \;=\; \sum_{E \in H_k} \rho(E),$$

which is a non-negative real number measuring the accumulated ethical weight of the history of exclusions that produced X.

The ordering $\lambda \leq \mu$ reflects the fact that principled refusal carries at least as much weight as instrumental commitment: to refuse on principle is to take a stronger stand than merely to choose one option over another. **Bind** events carry zero responsibility weight because they represent structural impossibility rather than agency; the system did not choose against those options, and it would be wrong to hold it answerable for their exclusion.

The total responsibility $\mathcal{R}(X)$ provides a second scalar measure on histories that runs parallel to the entropy measure $S(X)$. Both are derived from the same contraction of Ω_t, but they answer different questions: $S(X)$ measures the total magnitude of the contraction, while $\mathcal{R}(X)$ measures the portion of that contraction for which the system is answerable. The relationship between these two measures characterises the degree to which the history of an object was actively constructed by its agent as opposed to passively imposed by its environment.

22 Mechanics of Commitment

The axioms establish what can and cannot happen in the calculus, but they say nothing about the *cost* of building a world—the quantitative relationship between the freedom that is lost when a possibility is excluded and the actuality that is gained when it is committed. To capture this we introduce a discrete variational mechanics.

Definition 22.1 (Local Lagrangian)**.** At each step t, the local Lagrangian is

$$\mathcal{L}_t \;=\; \Delta\Omega_t \;+\; \lambda\,\Delta C_t,$$

where $\Delta\Omega_t = |\Omega_t| - |\Omega_{t+1}|$ is the reduction in optionality at step t and ΔC_t is the accounting cost—the ethical weight of the exclusion—weighted by a coupling constant $\lambda \geq 0$.

Definition 22.2 (Action)**.** The total action along a history $H = (x_1, \ldots, x_n)$ is

$$S[H] \;=\; \sum_{t=1}^{n} \mathcal{L}_t.$$

The action $S[H]$ is strictly non-decreasing along any admissible history: by Axiom **??** the option space can only shrink, so $\mathcal{L}_t \geq 0$ at every step.

Definition 22.3 (Commitment Momentum)**.** The commitment momentum π_t measures the rate at which future freedom is being converted into settled actuality:

$$\pi_t \;=\; -\frac{\partial \mathcal{L}_t}{\partial \Omega_t}.$$

Definition 22.4 (Hamiltonian)**.** The Hamiltonian $\mathcal{H}_t$ represents the agent's remaining maneuverability—the extent to which its future is still open:

$$\mathcal{H}_t \;=\; \pi_t\,|\Omega_t| \;-\; \mathcal{L}_t.$$

A system that never applies **Pop** or **Refuse** remains at maximal $\mathcal{H}_t$ throughout its existence. It is free in the most abstract sense—and inhabits no world in any concrete sense. Freedom without commitment is not depth; it is weightlessness, and weightlessness is not a form of richness but a form of absence.

The relationship to the entropy measures of Section 5 is direct. Each application of

$\mathbf{Pop}_x$ reduces $|\Omega_t|$ by one, decreasing the entropy $\log|\Omega_t|$ of the option space while increasing the Kolmogorov complexity of the commitment history. The Lagrangian makes this directional conversion quantitative: worldbuilding trades possibility for structure, and the action $S[H]$ measures the total cost of that trade along a given history.

23 The End of Philosophical Neutrality

The formal structures developed in the preceding sections treat irreversibility as a geometric property of histories. Hans Jonas forces a stronger conclusion: irreversibility is also an ethical property, and the two cannot be cleanly separated.

Jonas's central argument, developed across *The Phenomenon of Life* and *The Imperative of Responsibility*, is that philosophy cannot occupy a position of neutral observation with respect to the world it describes. To know is not to stand outside events and register them from a safe distance; to know is to participate in the events one describes, and participation carries obligation. Knowledge is always already ethical implication. The philosopher is not an external observer; she is always already involved, always already affected by what she studies, and that affectation is itself part of the responsibility she has assumed. The neutrality that the modern tradition aspired to—the view from nowhere, the disinterested gaze—is not available.

This argument has a direct formal consequence for the Spherepop calculus, and it is not imported from outside the system but follows from the logic of the axioms themselves.

Axiom 23.1 (Ethical Non-Neutrality)**.** Every act of exclusion within an admissible Spherepop process carries a non-zero responsibility weight:

$$\forall x \in H_t, \quad \rho_x \neq 0.$$

The Axiom of Ethical Non-Neutrality is not an additional decoration placed on top of the formal system. It is what Axiom **??** requires once one considers its implications carefully. If an exclusion is genuinely irreversible—if it truly and permanently forecloses a possibility—then the system that performed the exclusion is bound to it in a way that cannot be discharged by any subsequent operation. The responsibility weight ρ_x is the formal trace of that bindingness.

Remark 23.2. A system that treats all responsibility weights as zero is in effect claiming

that its exclusions are cost-free—that the possibilities it forecloses carry no weight that must be carried forward. But this claim is inconsistent with Axiom **??**: if an exclusion is irreversible, then the counterfactual world in which that exclusion was not made is permanently unavailable, and the system bears whatever consequences flow from that unavailability. The zero-weight assumption and the irreversibility axiom cannot both be held. A system that excludes without responsibility is not performing genuine irreversible exclusion; it is simulating exclusion while remaining indifferent to it.

Definition 23.3 (Accountable History)**.** An accountable history is a commitment history in which every appended event is accompanied by its responsibility weight:

$$H_t \;:=\; \big((x_1, \rho_{x_1}),\ (x_2, \rho_{x_2}),\ \ldots, (x_n, \rho_{x_n})\big).$$

Definition 23.4 (Accounting Functor)**.** The Accounting Functor

$$\mathbf{A} : \mathbb{P}(\Omega) \;\longrightarrow\; \mathcal{C}_{\text{cost}}$$

maps each history to its cumulative annotated record of why each exclusion occurred, distinguishing **Pop** (mechanical exclusion) from **Refuse** (principled exclusion) by the annotation attached to each event. It is a functor because the cost of a concatenated history is the composition of the costs of its parts.

Theorem 23.5 (Identity via Refusal)**.** *An agent's identity class is uniquely determined, up to observational equivalence under* **Collapse***, by its refusal set—the collection of events to which* **Refuse** *has been applied throughout its history.*

Sketch. Two agents with distinct refusal sets have distinct ethical annotations in their accountable histories. Since **A** is a functor, it maps distinct annotated histories to distinct objects in $\mathcal{C}_{\text{cost}}$. Two agents may have the same **Collapse** image—the same observable state—while having distinct refusal sets, but they remain intensionally distinct under **A**. □

Identity is forged as much by what an agent refuses as by what it commits to. An agent that commits to everything and refuses nothing has, in a precise sense, no more character than an agent that commits to nothing: both have failed to create the structure of principled exclusion that constitutes a self capable of maintaining integrity across time.

24 The Global Structure: Trajectory, History, and State

The development thus far has introduced three distinct but related levels of description, each arising from the previous by a projection that discards information. At the most fundamental level lie trajectories in the category $\mathcal{T}$ of possibility-history pairs, encoding the progressive contraction of admissible events and the full geometry of option-space dynamics. At the next level lie event histories in the Free History Category $\mathbb{P}(\Omega)$, obtained by projecting trajectories onto their irreversible commitment records. At the surface level lie observable states in a category $\mathcal{S}$, reconstructed from histories by composing the associated transformations. The present section unifies these levels into a single categorical structure and establishes the formal properties of the projections that connect them.

24.1 The Three Levels and Their Projections

The two projections have been defined in earlier sections. The first,

$$\Pi_{\text{hist}} : \mathcal{T} \longrightarrow \mathbb{P}(\Omega),$$

is the history projection of Section 16, which forgets the evolution of the option space and retains only the ordered commitment record. The second,

$$\textbf{Collapse} : \mathbb{P}(\Omega) \longrightarrow \mathcal{S},$$

is the Collapse functor of Section 9, which reconstructs an observable state from a history by composing the transformations associated with each committed event from a fixed root.

Both are functors. Π_{hist} respects composition because extending two trajectories in sequence produces the concatenation of their event words, which is composition in $\mathbb{P}(\Omega)$. **Collapse** respects composition because replaying a concatenated history is the same as composing the replays of its parts.

24.2 The Global Commutative Diagram

The composition of the two functors defines a map from trajectories directly to observable states:

$$\mathbf{Collapse} \circ \Pi_{\text{hist}} : \mathcal{T} \longrightarrow \mathcal{S}.$$

The global structure of the framework is expressed by the following diagram:

$$\begin{tikzcd}
\mathcal{T} \arrow[r, "\Pi_{\text{hist}}"] \arrow[dr, "\mathbf{Collapse}\circ\Pi_{\text{hist}}"'] & \mathbb{P}(\Omega) \arrow[d, "\mathbf{Collapse}"] \\
& \mathcal{S}
\end{tikzcd}$$

Proposition 24.1 (Commutativity of the global diagram)**.** *For every trajectory $\tau \in \mathcal{T}$,*

$$\mathbf{Collapse}\big(\Pi_{\text{hist}}(\tau)\big) \;=\; (\mathbf{Collapse} \circ \Pi_{\text{hist}})(\tau).$$

Proof. Immediate from the definition of functor composition. □

The diagram expresses a fundamental asymmetry: all observable states arise from histories, and all histories arise from trajectories, but neither of the reverse constructions is available. The arrows flow in one direction only, from the richest description to the coarsest, and the information lost at each step cannot be recovered from the output.

24.3 Non-Existence of Right Inverses

The irreversibility of the projections is expressed precisely by the non-existence of right inverse functors.

Theorem 24.2 (Irreversibility of projections)**.** *There exists no functor $\mathcal{S} \to \mathbb{P}(\Omega)$ that is a right inverse to* **Collapse***, and no functor $\mathbb{P}(\Omega) \to \mathcal{T}$ that is a right inverse to Π_{hist}.*

Proof. A right inverse to **Collapse** would be a functor $R : \mathcal{S} \to \mathbb{P}(\Omega)$ satisfying $\mathbf{Collapse} \circ R = \mathrm{id}_{\mathcal{S}}$. For every state $s \in \mathcal{S}$, the functor R would have to select a unique history H with $\mathbf{Collapse}(H) = s$. But as established in the reconstruction analysis of Section 15 and formalised in the categorical treatment of Section 16, multiple distinct histories can produce the same observable state. Any selection would have to

introduce additional information not present in s, contradicting the requirement that R be a functor from $\mathcal{S}$ alone.

A right inverse to Π_{hist} would be a functor $Q : \mathbb{P}(\Omega) \to \mathcal{T}$ satisfying $\Pi_{\text{hist}} \circ Q = \text{id}_{\mathbb{P}(\Omega)}$. For every event word $H \in \mathbb{P}(\Omega)$, the functor Q would have to produce a canonical trajectory τ with $\Pi_{\text{hist}}(\tau) = H$. But as shown in Proposition 16.3, multiple distinct trajectories project to the same event word, because the intermediate option-space structure is not encoded in H. Selecting a canonical trajectory would require information about which possibilities were available but not taken—information that Π_{hist} has discarded and that is not available from H alone. □

This theorem is the formal statement that you cannot recover history from state, and cannot recover possibility from history. Each projection discards information that is genuinely gone. The diagram has no section; the arrow of commitment is one-way throughout.

24.4 Entropy and Responsibility as Path Functionals

The entropy measure S and the responsibility measure $\mathcal{R}$ introduced in Sections 5 and 21 are defined on trajectories and measure properties of the full dynamics in $\mathcal{T}$. They are therefore path functionals on the deepest level:

$$S : \mathcal{T} \to \mathbb{R}_{\geq 0}, \qquad S(\tau) = \log \frac{|\Omega_0|}{|\Omega_k|},$$

$$\mathcal{R} : \mathcal{T} \to \mathbb{R}_{\geq 0}, \qquad \mathcal{R}(\tau) = \sum_{j=1}^{k} \rho(E_{i_j}).$$

Neither measure factors through Π_{hist} in general.

Proposition 24.3 (Non-factorization of path measures)**.** *There exist trajectories $\tau \sim \tau'$ such that $\Pi_{\text{hist}}(\tau) = \Pi_{\text{hist}}(\tau')$ but $S(\tau) \neq S(\tau')$ and $\mathcal{R}(\tau) \neq \mathcal{R}(\tau')$.*

Proof. As in the proof of Proposition 16.3, consider two trajectories that commit the same events in the same order but differ in their intermediate option spaces. If $|\Omega_k|$ differs between the two trajectories—because one was subject to additional **Bind** constraints that reduced the residual option space without committing events—then $S(\tau) = \log(|\Omega_0|/|\Omega_k|)$ takes different values. Similarly, if the two trajectories attribute

different responsibility weights to the same event sequence—one assigning some events as **Refuse** type and the other assigning them as **Pop** type—then $\mathcal{R}$ differs even though Π_{hist} maps both to the same event word. $\square$

This result establishes that entropy and responsibility refine the ontology beyond identity: they are properties of the full trajectory that survive the quotient $\sim$ only partially. Two objects with the same identity may nevertheless differ in the entropy cost and responsibility attribution of the trajectories that produced them. This does not undermine the identity criterion; it locates entropy and responsibility at the appropriate level, which is the level of trajectories rather than the level of event words.

24.5 Stratified Ontology

The global diagram, together with the path measures, reveals a stratified ontological structure with three levels and two associated scalar measures. The deepest level is the trajectory category $\mathcal{T}$, which carries the full geometry of possibility: the shapes of option spaces, the provenance of exclusions, and the measures S and $\mathcal{R}$ that quantify those properties. The intermediate level is the Free History Category $\mathbb{P}(\Omega)$, which carries the ordered record of committed events and supports the identity criterion. The surface level is the observable state category $\mathcal{S}$, which carries only the present configuration derivable from the history without further reference to its internal structure.

Information is conserved at the trajectory level and progressively lost at each subsequent level. The projections are the formal instruments of that loss. The identity of an object is what remains after the first projection; the observable state of an object is what remains after both. Entropy and responsibility are shadows of the trajectory that neither projection fully captures, but that leave traces at the levels of history and state through the specific event words and state values they constrain.

The full structure may be visualised as a tower:

$$\begin{tikzcd}
\mathcal{T} \arrow[d, "\Pi_{\text{hist}}"'] \arrow[dd, bend left, "\textbf{Collapse}\circ\Pi_{\text{hist}}"] \\
\mathbb{P}(\Omega) \arrow[d, "\textbf{Collapse}"'] \\
\mathcal{S}
\end{tikzcd}$$

with S and $\mathcal{R}$ living at the top and partial information about them propagating downward through the functors.

25 Worldhood and Agency

The stratified ontology of Section 24 defines three levels of description connected by irreversible projections. Every system that can be described within this framework occupies some position in the tower, and all systems produce histories that can be compared and measured. But there is a crucial distinction that the tower does not yet capture: the distinction between systems that genuinely participate in the dynamics of possibility-space contraction and systems that merely sample from a fixed pool of outputs. The present section formalises this distinction as the criterion of worldhood and shows that it is equivalent to possessing a non-trivial Spherepop identity.

25.1 Commitment as Strict Contraction

Recall that the evolution of a Spherepop system traces a path through states (Ω_t, H_t) subject to the monotone contraction condition $\Omega_{t+1} \subseteq \Omega_t$. This condition allows for transitions in which the option space shrinks strictly and transitions in which it remains the same—the latter corresponding to **Bind** operations that impose structural constraints without appending to the history.

We define commitment to require strictly more: a genuine reduction in the space of future possibilities as a direct consequence of the system's own action.

Definition 25.1 (Commitment)**.** A transition $(\Omega_t, H_t) \to (\Omega_{t+1}, H_{t+1})$ is a *commitment* if the history grows and the option space shrinks strictly: $H_{t+1} = H_t \cdot E$ for some event E, and $\Omega_{t+1} \subsetneq \Omega_t$ with $E \notin \Omega_{t+1}$.

Commitment is therefore strictly more than state transition. A system that transitions between states without committing is one that changes its observable output without foreclosing any possibility. Its history grows formally—it appends events to its word—but those events leave the option space unchanged, which means that, at the level of the trajectory category, the extension carries no genuine contraction of the future.

25.2 Worldhood

Definition 25.2 (Worldhood)**.** A system exhibits *worldhood* if its evolution contains an unbounded sequence of commitments: there exists a sequence of stages $t_1 < t_2 < \cdots$ such that each transition at those stages is a commitment.

A world-inhabiting system is one that continually carves a path through an initially undifferentiated future, making each commitment irrevocable and accumulating an ever-growing history of irreversible exclusions. Worldhood is not a property of any snapshot of the system but of the ongoing process of its evolution.

25.3 Simulators

The complementary class consists of systems that produce outputs without committing.
Definition 25.3 (Simulator)**.** A system is a *simulator* if $|\Omega_{t+1}| = |\Omega_t|$ for all transitions in its evolution.

A simulator may generate arbitrarily complex sequences of events. It may produce outputs that are statistically indistinguishable from those of a world-inhabiting system across any finite observation window. But its option space does not contract, so no commitment ever occurs, and the history it accumulates is a formal record of outputs rather than an irreversible trace of foreclosures. The distinction between a simulator and a world-inhabiting system is not observable from the outputs alone; it is a property of the trajectory in $\mathcal{T}$.

25.4 The Worldhood Theorem

Theorem 25.4 (Worldhood Theorem)**.** *A system admits a non-trivial identity in the Spherepop sense if and only if it exhibits worldhood.*

Proof. ($\Rightarrow$): Suppose a system admits a non-trivial Spherepop identity. Then there

exist at least two distinct objects it can produce, corresponding to distinct event words $H \neq H'$ in $\mathbb{P}(\Omega)$. These event words arise as images under Π_{hist} of distinct trajectories, and the existence of distinct event words requires that the histories differ at some stage—that is, that at some stage a distinct event was committed. Each such commitment involves a strict contraction $\Omega_{t+1} \subsetneq \Omega_t$. Since distinct event words require differences at infinitely many potential stages (because the free monoid $\mathcal{E}^*$ is infinite), the system can produce an unbounded sequence of distinct commitments, establishing worldhood.

($\Leftarrow$): Suppose the system exhibits worldhood, so that there is an unbounded sequence of commitments at stages $t_1 < t_2 < \cdots$. Each commitment appends an event to the history and strictly contracts the option space. The sequence of event words $H_{t_1}, H_{t_2}, \ldots$ is strictly increasing under the prefix ordering in $\mathbb{P}(\Omega)$, with each subsequent word being a strict extension of all prior ones. These words are therefore distinct elements of $\mathbb{P}(\Omega)$, and the objects they define are distinct under the Spherepop identity criterion. The system therefore admits a non-trivial identity structure. □

The theorem establishes that identity and commitment are inseparable in the Spherepop framework. A system that does not reduce its option space cannot produce a growing commitment history in the relevant sense; any event words it generates are formal labels rather than traces of foreclosure. Conversely, a system that genuinely contracts its option space accumulates a history that is not only formally recorded but irreversible in the ontological sense: the foreclosed possibilities are gone, and the event word that records their foreclosure is the only residue of what they were.

25.5 Agency as Structured Commitment

Worldhood requires commitment but does not yet distinguish between systems whose commitments arise from their own agency and systems whose commitments are entirely environmentally imposed. The responsibility measure introduced in Section 21 enables this further distinction.

Definition 25.5 (Agency)**.** A system exhibits *agency* if its total responsibility measure $\mathcal{R}(\tau) = \sum_j \rho(E_{i_j})$ is strictly positive and not determined solely by the entropy weights $w(E_{i_j})$.

An agent is a world-inhabiting system that is answerable for some portion of its option-

space contractions—one for which at least some exclusions carry **Pop** or **Refuse** type annotations rather than being entirely the product of structural **Bind** constraints imposed from outside. The distinction matters because identity requires worldhood while accountability—in the ethical and legal senses—requires agency: it requires that the contractions which define the system's history are not merely suffered but in part performed.

The relationship between entropy and responsibility can now be made precise. Both measures are path functionals on $\mathcal{T}$, and both are non-negative. Entropy measures the total magnitude of option-space contraction; responsibility measures the portion of that contraction attributable to the system's own action. A system with high entropy but zero responsibility is one whose option space contracted entirely due to external constraints; it has a rich history but is answerable for none of it. A system with non-zero responsibility has committed at least some of its history actively, and its identity is constituted at least in part by what it chose to foreclose.

25.6 The LLM Boundary Case

The Worldhood Theorem has an immediate and sharp application to large language models. A language model produces text by sampling from a probability distribution conditioned on a context. After each generation step, the model's parameters are unchanged, its capacity to sample from the same distribution is undiminished, and the probability distribution it faces at the next step is of comparable richness to the one it faced at the previous step. In the language of the framework: $|\Omega_{t+1}| = |\Omega_t|$ at every step. The model is a simulator.

Corollary 25.6 (Language models and worldhood)**.** *A language model operating by autoregressive sampling does not exhibit worldhood and therefore does not admit a non-trivial Spherepop identity.*

This is not a criticism of language models as useful computational systems. It is a formal statement about what kind of entity they are. They produce event sequences—tokens—but those sequences are not irreversible boundary traces of a contracting possibility space; they are selections from an invariant distribution. The model's "history" in the sense of its context window is a record of outputs, not an accumulation of foreclosures. It can represent histories but does not have one in the relevant sense.

The corollary makes precise the intuition that hallucination in language models is not

primarily an accuracy problem but an ontological one. A hallucination is an output produced without any cost in foreclosed possibility: the model generates a confident assertion without eliminating any alternative from its option space, and therefore without bearing any responsibility for the exclusion. The output is meaningful at the level of $\mathcal{S}$—it has propositional content—but it has no history at the level of $\mathbb{P}(\Omega)$ and no trajectory at the level of $\mathcal{T}$. It is a state without a committed history behind it.

26 Artificial Intelligence as a Boundary Case

Large language models present an instructive boundary case. A language model produces text by sampling from a probability distribution conditioned on a context. This act of sampling does not reduce the model's internal option space in any sense that the Spherepop calculus recognises: its parameters are unchanged after each generation, its capacity to sample from the same distribution is fully intact, and at the next step it will face a probability distribution of comparable richness. The model generates output, but it does not commit.

Definition 26.1 (Zero-Cost Structure)**.** A system exhibits *zero-cost structure* if $\Delta\Omega_t = 0$ for all t: its generative acts do not reduce its option space and therefore do not accumulate commitment momentum or contribute to an accountable history.

Corollary 26.2 (Simulation Without Worldhood)**.** *Any system with non-shrinking option space satisfies $\pi_t = 0$ for all t and cannot instantiate a history-defined identity or inhabit a world in the sense of Theorem 25.4.*

Proof. $\pi_t = -\partial\mathcal{L}_t/\partial\Omega_t$. If $\Delta\Omega_t = 0$, the geometric term in the Lagrangian vanishes. With no reduction in optionality, the commitment momentum is zero. By Axiom 23.1, a system with zero commitment momentum carries no non-zero responsibility weights and therefore has no accountable history. □

The diagnosis of hallucination follows. A hallucination is structure generated without the cost of permanent exclusion: the model produces a confident assertion without having foreclosed any alternative, and therefore without bearing responsibility for the foreclosure. The output is meaningful to the user who reads it, but it carries no weight for the system that produced it, because it does not constrain that system's future behaviour. Hallucination is not primarily an engineering failure but a structural feature of a system whose ontological type is generator rather than agent: it speaks without

standing by what it says, produces histories in the syntactic sense while accumulating no accountable history in the semantic sense.

This is not a criticism of language models as tools. It is a formal statement about what kind of entity a language model is—a statement that has consequences for questions about commitment, identity, and responsibility, and that the Worldhood Theorem makes precise.

27 Stratified Architecture and Non-Interference

The formal development has architectural consequences for any implementation of the Spherepop calculus. The most important is the Non-Interference Theorem: the authoritative causal record must be shielded from any influence by the observational projections derived from it.

The architecture is stratified into three layers. The *Calculus Layer* $\mathcal{C}_{\text{calc}}$ is the abstract category of Spherepop morphisms, providing meaning, typing, and ethical annotation. The *Kernel Layer* is the deterministic interpreter $\mathbf{K} : \mathcal{C}_{\text{calc}} \to \mathbb{P}(\Omega)$, which maintains the append-only log H_t as the sole authoritative source of causal truth. Once an event has been committed to the log it cannot be retracted, amended, or reinterpreted. The *View Layer* is the projection $\mathbf{V} : \mathbb{P}(\Omega) \to \mathcal{V}$, which derives observational snapshots and speculative branches from the canonical history without contaminating it.

Theorem 27.1 (Non-Interference)**.** *There is no functor $\mathcal{V} \to \mathbb{P}(\Omega)$ that feeds View-layer data back into the Kernel.*

Proof. Such a functor would require View-layer objects—which may be speculative, reordered, or filtered—to produce authoritative extensions of the canonical history H_t. But H_t is governed by Axiom **??**, which requires it to be append-only, and $\mathbb{P}(\Omega)$ contains no non-identity isomorphisms (since prefix ordering is antisymmetric). A functor from $\mathcal{V}$ to $\mathbb{P}(\Omega)$ would have to either produce append-only outputs from inputs that may not be append-only, which is impossible in general, or introduce inverse morphisms into $\mathbb{P}(\Omega)$, which contradicts the absence of non-identity isomorphisms. □

$$\begin{tikzcd} \mathcal{C}_{\text{calc}} \arrow[r, "\mathbf{K}"] & \mathbb{P}(\Omega) \arrow[r, "\mathbf{V}"] & \mathcal{V} \arrow[ll, dashed, bend left, "\nexists"] \end{tikzcd}$$

The dashed arrow is the prohibited feedback path. The philosophical import is that observation cannot rewrite the past. A View is a perspective on history, not a revision of it. If speculative reasoning in the View Layer could feed back into the Kernel and alter the canonical record, the identity of the system—which just is that record—would become hostage to whatever perspective happened to be applied most recently.

28 A Variational Principle for Commitment

The preceding sections have been structural: they define trajectories, establish their projections, characterise identity as an equivalence class, and introduce measures that quantify the contraction of possibility. What remains is a generative principle. Rather than treating trajectories as arbitrary paths through the space $\mathcal{T}$ subject only to the irreversibility axioms, we ask whether realised trajectories are in some sense optimal—extremals of a functional that encodes the competing demands of entropy, responsibility, and coherence.

This question has precedent in classical and quantum mechanics, where realised physical trajectories are characterised as extremals of an action functional, and the Euler-Lagrange equations express the local optimality conditions that every point of a realised trajectory must satisfy. The Spherepop analogue operates in the discrete setting of the trajectory category $\mathcal{T}$, but the structural logic is the same: among all possible sequences of exclusions connecting the same initial and terminal conditions, realised histories are those that extremize a cumulative cost.

28.1 The Action Functional

Let $\tau : (\Omega_0, \varepsilon) \to \cdots \to (\Omega_k, H_k)$ be an admissible trajectory in $\mathcal{T}$. We define the action of τ by

$$\mathcal{A}(\tau) \;=\; \sum_{j=1}^{k} \mathcal{L}(\Omega_{j-1}, E_{i_j}),$$

where $\mathcal{L}$ is a discrete Lagrangian assigning a local cost to each transition in the trajectory.

The Lagrangian is taken to have the form

$$\mathcal{L}(\Omega, E) \;=\; \alpha\, w(E) \;+\; \beta\, \rho(E) \;+\; \gamma\, \Phi(\Omega, E),$$

where $w(E) = \log(|\Omega|/|\Omega'|)$ is the entropy weight measuring the magnitude of the option-space contraction at this step, $\rho(E)$ is the responsibility weight measuring the provenance of the exclusion, and $\Phi(\Omega, E)$ is a coherence functional measuring how the elimination of event E fits with the structure of the remaining option space after its removal. The non-negative constants α, β, γ determine the relative importance of each term.

28.2 Interpretation of the Three Terms

The three terms in the Lagrangian correspond to three distinct aspects of the commitment represented by each transition.

The entropy term $\alpha\, w(E)$ measures the magnitude of foreclosure at each step. A transition that eliminates a large fraction of the remaining option space carries a high entropy cost. When α is large relative to the other constants, trajectories are penalised for making large eliminations at single steps, and the extremal principle favours histories that spread their foreclosures evenly across many small steps rather than concentrating them in a few large ones. This is the sense in which entropy enters not just as a measure but as a constraint on how commitment unfolds over time.

The responsibility term $\beta\, \rho(E)$ measures the attribution of each foreclosure. When β is large, the extremal principle distinguishes sharply between eliminations that are structurally imposed (**Bind** events with $\rho = 0$) and those that are actively chosen or principled (**Pop** and **Refuse** events with $\rho > 0$). Trajectories with high total responsibility are more "costly" in the Lagrangian sense, which means that realised trajectories under a high-β regime are those that minimise the number or weight of active commitments relative to structural constraints. This is the formal counterpart of the observation that agency— genuine choice—is costly in a way that structural impossibility is not.

The coherence term $\gamma\, \Phi(\Omega, E)$ measures the compatibility of each elimination with the structure of what remains. A trajectory that eliminates possibilities in an order that preserves the internal consistency of the remaining option space—that does not fragment it into disconnected components or introduce contradictions between the remaining alternatives— carries lower coherence cost than one that leaves the option space structurally incoherent. The coherence term formalises the intuition that well-constructed worlds have a kind of internal logic: each commitment is compatible with

the remaining structure, and the trajectory does not introduce gratuitous disorder into what it leaves behind.

28.3 The Principle of Stationary Commitment

Axiom 28.1 (Stationary Commitment)**.** Realised trajectories in an admissible Sphere-pop system are stationary points of the action functional $\mathcal{A}$ under admissible variations: if τ is a realised trajectory and τ' is any admissible trajectory with the same initial and terminal states, then $\mathcal{A}(\tau) \leq \mathcal{A}(\tau')$.

An admissible variation is a modification of the sequence of eliminations that preserves the initial state (Ω_0, ε) and terminal event word H_k while potentially altering the intermediate option-space structure and the ordering of eliminations at steps where events are causally independent.

The principle does not uniquely determine the realised trajectory—there may be multiple extremals, and the set of admissible variations may be large—but it identifies a class of trajectories as physically or computationally preferred: those that minimise the total cost of constructing the world they construct.

28.4 The Discrete Euler-Lagrange Condition

From the stationarity condition, a local optimality requirement emerges for each step of a realised trajectory.

Proposition 28.2 (Discrete Euler-Lagrange condition)**.** *If τ is a stationary trajectory, then for each step j and each admissible alternative event $E' \in \Omega_{j-1}$ that could have been committed at that step in place of E_{i_j}, the realised event satisfies*

$$\mathcal{L}(\Omega_{j-1}, E_{i_j}) \;\leq\; \mathcal{L}(\Omega_{j-1}, E').$$

Proof. If there were an alternative event E' with strictly lower Lagrangian cost at step j, and if the resulting trajectory after substituting E' for E_{i_j} remained admissible and reached the same terminal event word (possibly after reordering causally independent subsequent events), then the original trajectory would not be a minimum of $\mathcal{A}$. The stationarity assumption therefore requires that the realised event minimises the local Lagrangian cost over all admissible alternatives at each step. □

The local optimality condition is the discrete analogue of the Euler-Lagrange equations: each commitment is locally least-cost given the current option space and the Lagrangian weighting. A trajectory composed of locally optimal steps is globally stationary when no sequence of local substitutions can reduce the total action.

28.5 The Arrow of Time from Action Accumulation

The variational formulation provides an intrinsic account of temporal direction that does not need to be postulated separately. Because $w(E) \geq 0$ and $\rho(E) \geq 0$ for all events, the Lagrangian $\mathcal{L}(\Omega, E)$ is non-negative for all Ω and E. It follows that the action accumulates monotonically along any admissible trajectory:

$$\mathcal{A}(\tau_{0\to t'}) \;\geq\; \mathcal{A}(\tau_{0\to t}) \quad \text{for all } t' > t.$$

Time, in this framework, is the accumulation of action. The arrow of time is not imposed as an external condition on the dynamics but emerges from the non-negativity of the Lagrangian: commitment always costs something, and that cost can only accumulate, never decrease. This is the variational expression of the same irreversibility that the axioms of Section 4 encode structurally.

28.6 Relationship to the Entropy and Responsibility Measures

The action functional $\mathcal{A}$ subsumes both the entropy measure S and the responsibility measure $\mathcal{R}$ as special cases determined by the coupling constants. When $\beta = \gamma = 0$, the action reduces to $\mathcal{A}(\tau) = \alpha \cdot S(\tau)/\log$, which is proportional to the total entropy of option-space contraction. When $\alpha = \gamma = 0$, it reduces to $\mathcal{A}(\tau) = \beta \cdot \mathcal{R}(\tau)$, which is proportional to total responsibility.

More generally, $\mathcal{A}(\tau)$ is a weighted combination of these two measures plus the coherence term, and the extremal trajectories are those that balance all three. A system with large α constructs worlds by spreading its foreclosures evenly. A system with large β constructs worlds by minimising the number of active choices, delegating as much as possible to structural constraints. A system with large γ constructs worlds that maintain internal coherence at each step. The coupling constants are therefore not merely numerical parameters but characterise the style or disposition of the world-constructing

system.

28.7 Relationship to the Global Categorical Diagram

The variational principle operates at the level of the trajectory category $\mathcal{T}$ and does not directly constrain the projections Π_{hist} or **Collapse**. However, it does constrain the set of trajectories that appear in the diagram: among all trajectories that project to the same event word H under Π_{hist}, the stationary principle selects those that minimise the action among all trajectories in the preimage $\Pi_{\text{hist}}^{-1}(H)$.

This means that the identity of an object—its equivalence class $[\tau]$ in $\mathcal{T}$—is not just any class of trajectories that happen to produce the same event word. It is, in the variational reading, the class of *optimal* trajectories that produce that word: those for which no admissible alternative with the same boundary conditions carries lower action. The history of an object is therefore the trace of an extremal process, and the event word that encodes its identity is the image of that extremal under Π_{hist}.

28.8 Interpretation

The variational principle completes the framework by providing a generative law for world-construction. Trajectories are not merely arbitrary sequences of exclusions satisfying the irreversibility axioms; they are solutions to an optimisation problem that balances the cost of entropy, the cost of responsibility, and the maintenance of coherence.

From this perspective, the history of an object is not a record of arbitrary past decisions but a record of the decisions that, given the initial conditions and the Lagrangian weighting, were the locally optimal ones. Every event in the history was the best available act at the moment it occurred, in the sense determined by the Lagrangian. Identity is therefore not merely the shadow of an irreversible process; it is the shadow of an *optimal* irreversible process, one that could not have been otherwise at any step without incurring higher total cost.

The connection to physical principles is direct. In classical mechanics, the principle of stationary action selects realised trajectories from the space of all possible paths. In quantum mechanics, the path integral assigns amplitudes to all paths and selects the dominant ones by interference. The Spherepop variational principle occupies an

analogous position in the theory of identity-constituting processes: among all ways of constructing an object, the realised way is the stationary one, and the identity of the object is the invariant of that realised construction.

The world is not simply what happened; it is what happened optimally. And identity is the record of that optimality, compressed into the irreversible boundary trace of the process that produced it.

29 Conclusion: From Membership to Commitment

The framework developed in this essay proposes a fundamental reorientation of how identity is to be understood in mathematics, computation, and ontology. The prevailing paradigm, inherited from set theory, locates identity in membership: an object is what it contains, and two objects are identical if their contents coincide. This criterion is extensional, static, and atemporal. It treats the past as irrelevant and the future as unconstrained, reducing identity to a snapshot that discards the process by which that snapshot came to be. Two datasets with identical contents are the same dataset; two organisms with the same genetic sequence are the same organism; two computations that produce the same output have the same identity as programs. In each case, the history is a biographical footnote rather than a constitutive fact.

The Spherepop framework replaces this criterion with one grounded in irreversibility. An object is not defined by what it contains but by what has happened to produce it. Identity is not a property of a state but of a trajectory, and it is preserved not by structural equivalence but by the equality of irreversible histories. The event word, obtained as the projection of a trajectory through possibility space, serves as the canonical representative of this identity: minimal in that it forgets all inessential variation in the trajectory, and complete in that no further distinctions can be recovered once the projection has been applied.

This shift from membership to commitment has several consequences that run throughout the formal development.

First, it renders time constitutive rather than incidental. The order of events is not an annotation on identity but its substance. Two computations that produce the same output by different orderings of independent events are the same object under extensional identity and distinct objects under Spherepop identity. The Necessity Theorem

(Theorem 17.3) establishes that this is not merely a design choice: under genuine irreversibility, the extensional criterion fails in both directions—conflating some distinct objects and failing to cohere with others—so that historical identity is the only consistent alternative. The shift from membership to commitment is therefore not optional. It is what irreversibility requires.

Second, it replaces the static ontology of sets with a dynamic ontology of processes. Objects are not the residents of abstract containers; they are the terminal nodes of trajectories through spaces of diminishing possibility. The container metaphor asks: what is this object made of? The trajectory metaphor asks: what path of foreclosures produced this object? The first question permits a timeless answer. The second requires a history. The introduction of the state pair (Ω_t, H_t) and the possibility preimage construction of Section 4 translate the trajectory metaphor into precise formal terms: identity as event history is the projection Π_{hist} of the dynamical process, and the full content of an object is visible only in the trajectory that produced it.

Third, it introduces a natural hierarchy of projections that replaces the flat universe of sets with a stratified ontology. The global categorical diagram of Section 24 expresses this hierarchy precisely: the trajectory category $\mathcal{T}$ contains the maximal information, the Free History Category $\mathbb{P}(\Omega)$ retains the ordered record of commitment, and the observable state category $\mathcal{S}$ retains only the present configuration. Each projection discards information irreversibly. No functor reverses either projection. The stratification is not a bookkeeping device but an expression of the same asymmetry of time that motivates the entire framework.

Within this hierarchy, entropy and responsibility emerge as intrinsic measures of the same underlying process rather than externally assigned weights. Entropy quantifies the total magnitude of option-space contraction along a trajectory; responsibility quantifies the portion of that contraction attributable to the system's own agency rather than to structural constraints. Both are path functionals on $\mathcal{T}$; neither factors through Π_{hist} in general. Identity, entropy, and responsibility are therefore not independent features of the framework but aspects of a single structure: the irreversible elimination of alternatives, measured from different angles.

The Worldhood Theorem (Theorem 25.4) sharpens this picture by identifying the class of systems that genuinely participate in the structure. A system that does not contract its option space cannot accumulate a meaningful history and therefore cannot possess

Spherepop identity. The Worldhood Theorem proves that this is an exact equivalence: worldhood is necessary and sufficient for non-trivial identity. Systems that lack worldhood—including any system that generates outputs by sampling from a fixed distribution—are simulators in the formal sense: they can represent the surface of the structure without participating in its dynamics. The outputs may be indistinguishable at the level of $\mathcal{S}$, but the underlying difference in trajectory type is categorial.

The variational principle of Section 28 provides the deepest layer: a generative law from which trajectories arise not as arbitrary sequences satisfying the irreversibility axioms but as solutions to an optimisation problem. The realised history of an object is the trace of a stationary trajectory—one that balances the cost of entropy, the weight of responsibility, and the maintenance of coherence at each step. Identity is therefore not merely the shadow of an irreversible process; it is the shadow of an optimal irreversible process. The event word records not just what happened but what could not have been otherwise given the initial conditions and the Lagrangian that governs the system's construction.

The resulting picture is one in which objects are made rather than given, constructed through irreversible acts of commitment rather than discovered as pre-existing structures. To exist as a Spherepop object is to have foreclosed alternatives. To have foreclosed alternatives is to have a history. To have a history is to have been produced by a trajectory that cannot be reconstructed from the present state alone. And to have been produced by a trajectory that extremizes a cost functional is to be the outcome of a process that could not, in the relevant sense, have been otherwise.

This perspective does not merely extend existing formalisms; it challenges their foundations. Set-theoretic identity works by asking what an object is at a given moment. Spherepop identity works by asking what trajectory of foreclosures produced the object. The first question yields an atemporal criterion. The second yields a criterion that is constitutively temporal, constitutively irreversible, and constitutively tied to the space of alternatives that the object's history has permanently closed.

In place of an ontology of membership, Spherepop proposes an ontology of commitment. In place of static equivalence, irreversible construction. In place of objects defined by what they contain, processes defined by what they have made impossible. The world is not a collection of containers; it is the residue of a history of foreclosures, and identity is the ordered record of that history—irreversible, incompressible without loss, and

prior to any observable present state.

30 A BNF Grammar for Spherepop Expressions

The abstract syntax of Spherepop expressions may be given precisely by a grammar in Backus–Naur Form (BNF). This grammar generates the symbolic representations of event histories that have been used throughout the main text, and provides a foundation for the rewriting system of Appendix 31.

The lexical primitives are identifiers for regions and events. Identifiers are sequences of letters and digits beginning with a letter.

```
<identifier> ::= <letter>
               | <identifier> <letter>
               | <identifier> <digit>

<letter>     ::= a | b | ... | z | A | B | ... | Z
<digit>      ::= 0 | 1 | 2 | ... | 9
```

Event names are identifiers equipped with a numeric subscript, representing the label of the event in the global enumeration.

```
<event>      ::= "E" <digit-seq>
<digit-seq>  ::= <digit> | <digit> <digit-seq>
```

An event word records the ordered sequence of events in a history. The empty event word, corresponding to the identity object with no generative history, is denoted ε.

```
<event-word> ::= <epsilon>
               | <event>
               | <event> "," <event-word>
```

The Spherepop expression of an object wraps its event word in the canonical notation introduced in Section 3.

```
<spherepop-expr> ::= "(" "spherepop" "(" <object> ")" ")"
<object>         ::= <identifier>
```

Branching and merging events are represented by the following productions, which correspond to the split and merge constructions discussed in Sections 2 and 10.

```
<split>  ::= "split" "(" <object> ")" "->" <object> "," <object>
<merge>  ::= "merge" "(" <object> "," <object> ")" "->" <object>
```

The full syntactic description of a history is a sequence of statements, each of which is either a split, a merge, or a direct event transformation.

```
<history>        ::= <statement>
                   | <statement> <history>
<statement>      ::= <split>
                   | <merge>
                   | <event-transform>
<event-transform>::= <object> "->" <event> "->" <object>
```

A normalized Spherepop expression is one in which the event word has been ordered according to the causal dependencies of the event graph, as described in Section 8. The grammar for normal forms therefore coincides with the grammar for event words, with the additional requirement that the ordering respects the partial order induced by the event graph.

```
<normal-form> ::= "(" <event-word> ")"
```

Two Spherepop expressions represent identical objects precisely when they normalize to the same normal form according to the rewriting rules of Appendix 31. The grammar therefore provides the syntactic substrate on which the normalization theory operates, and together they supply a decision procedure for Spherepop identity: parse the expression, construct the event graph, compute a topological ordering, and compare the resulting normal forms.

31 Rewriting Rules and Normalization

The normalization procedure described in Section 8 may be given a precise operational account in terms of a rewriting system acting on Spherepop expressions. Rewriting systems provide a standard framework for defining canonical forms through the iterated application of local rules, and their confluence and termination properties guarantee that the normalization procedure is well-defined and terminates [Abramsky and Jung, 1994].

The central rewriting rule exploits the independence relation $\parallel$ defined in Section 8.

When two adjacent events in an event word are independent, their order may be exchanged.

$$(E_i, E_j) \longrightarrow (E_j, E_i) \quad \text{whenever } E_i \parallel E_j.$$

This rule is the Spherepop analogue of the commutation relations that define Mazurkiewicz trace equivalence [Mazurkiewicz, 1987]. By applying this rule repeatedly in a prescribed order—for instance, by always commuting the leftmost independent pair—one obtains the normal form of the history.

In addition to commutation, the rewriting system includes rules for restructuring split and merge expressions. A merge expression $\text{merge}(A, B) \to D$ may replace any pair of incoming edges into the vertex representing D in the event graph, while a split expression $\text{split}(R) \to A, B$ may replace a pair of outgoing edges from R. These structural rules convert compound symbolic descriptions of event graphs into standardized forms that are amenable to the commutation rule.

The normalization algorithm proceeds in three conceptual phases. In the first phase, the symbolic expression is parsed and the corresponding event graph is constructed explicitly. Vertices are created for each region appearing in the expression, and directed edges are inserted for each event. In the second phase, a topological ordering of the event graph is computed; this is always possible since the graph is a DAG. In the third phase, commutation rules are applied to bring the event word into agreement with the topological ordering.

The resulting sequence is the Spherepop normal form.

Proposition 31.1 (Confluence)**.** *The commutation rewriting system is confluent: all rewriting paths from a given event word lead to the same normal form, regardless of the order in which commutation rules are applied.*

Confluence follows from the fact that the independence relation $\parallel$ is symmetric and the commutation rules satisfy the diamond property: whenever two distinct commutation rules are applicable to adjacent pairs in an event word, both can be applied in sequence, and the results join after a finite number of additional steps. The proof is analogous to the confluence proof for Mazurkiewicz traces [Mazurkiewicz, 1987] and will not be repeated here in full.

Confluence guarantees that the normal form is independent of the particular rewriting strategy used to compute it. Two Spherepop expressions represent the same historical object precisely when their rewriting sequences converge to the same normal form. The rewriting system thus provides an operational decision procedure for Spherepop identity that is both sound and complete with respect to the causal equivalence relation.

32 Causal Traces, Petri Nets, and String Diagram Representations

The event histories of Spherepop share structural features with three established mathematical frameworks for representing causal processes: trace theory, Petri nets, and string diagram formalisms. Examining these connections clarifies the position of Spherepop within the broader landscape of process-based formalisms and suggests directions for further technical development.

In the Mazurkiewicz trace theory of concurrency [Mazurkiewicz, 1987], a trace is an equivalence class of event sequences modulo the commutation of independent events. The event set $\mathcal{E}$ is equipped with an independence relation $\|$, and two sequences are identified when they are related by the exchange of adjacent independent events. The resulting equivalence classes—the traces—represent the causal content of an execution, abstracting away from the arbitrary serialization of concurrent actions. The Spherepop normal form construction is precisely the operation of selecting a canonical representative from each trace class. The commutation rules of Appendix 31 are the Spherepop instantiation of the Mazurkiewicz commutation relations, and the confluence result established there corresponds to the standard confluence of the trace rewriting system.

The connection to Petri nets [Petri, 1966] is equally direct. In a Petri net, *places* represent conditions or resources and *transitions* represent events that consume and produce tokens at places. A simple correspondence maps Spherepop regions to places and Spherepop events to transitions. Under this mapping, split events correspond to transitions with multiple output places (producing a token in each of several descendant regions), while merge events correspond to transitions with multiple input places (requiring a token in each of the regions being combined before the merged region can be produced). A Spherepop history then corresponds to a *firing sequence* in the Petri net: a sequence of enabled transitions that produces the final configuration from the

initial one. The DAG property of Spherepop event graphs corresponds to the property of the Petri net that the firing sequence contains no transition that could return the net to a previously visited marking.

String diagrams provide a geometric rather than algebraic language for compositional processes in monoidal categories [Baez and Stay, 2010]. In a string diagram, objects are represented by wires and morphisms are represented by boxes or nodes through which wires pass. Sequential composition is represented by connecting the output wire of one morphism to the input wire of the next; parallel composition is represented by placing wires side by side without connecting them. Spherepop events map naturally onto this language: a split event appears as a box with one input wire and two output wires, a merge event as a box with two input wires and one output wire, and a sequential transformation as a box with one input and one output. A complete Spherepop history becomes a compound string diagram assembled from these elementary components.

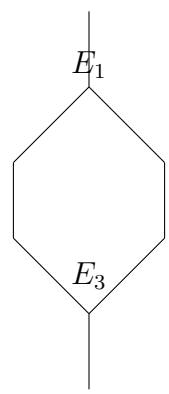

Figure 9: A string diagram representation of a split event E_1 followed by a merge event E_3. The single input wire at the top corresponds to the initial region R; the two intermediate wires carry the descendant regions A and B; and the single output wire at the bottom corresponds to the merged region D.

As illustrated in Figure 9, string diagrams make the compositional structure of event histories visually transparent. Sequential composition corresponds to vertical stacking of diagram components, while parallel processes correspond to horizontal juxtaposition. The diagram in Figure 9 corresponds to the simple history $(\text{spherepop}(D)) = (E_1, E_3)$, where E_1 is the split and E_3 is the merge. More complex histories are represented by more elaborate compound diagrams.

What distinguishes Spherepop from the standard monoidal category framework is its emphasis on *historical identity* as the primary ontological criterion. The string di-

agrams and Petri net representations provide operational and geometric pictures of the same underlying structure, while the event-word and normal-form apparatus provides a canonical symbolic encoding. Spherepop integrates these perspectives into a unified framework where causal structure, compositional irreversibility, and identity under historical comparison are treated as equally fundamental features.

33 Spherepop and Derived Causal Categories

The constructions developed in the main text suggest that Spherepop histories may be situated within a higher-categorical framework in which the causal categories of Section 9 are enriched by the rewriting operations of Appendix 31. This section sketches the relevant structures and their relationship to derived categories and sheaf theory, with the aim of indicating where more technical development could be pursued.

In the causal history category $\mathcal{H}$ introduced in Section 9, morphisms are irreversible events and their compositions are directed causal chains. The rewriting rules of Appendix 31 provide natural transformations between parallel event chains that differ only by the ordering of independent events. A commutation

$$(E_i, E_j) \Rightarrow (E_j, E_i) \quad \text{when } E_i \parallel E_j$$

may be interpreted as a 2-morphism in a bicategory or 2-category in which 1-morphisms are event chains and 2-morphisms are commutation steps. The confluence result of Appendix 31 then corresponds to the coherence of these 2-morphisms: any two sequences of commutation steps relating the same pair of event chains are connected by higher-order equivalences.

This higher-categorical structure suggests that the objects of Spherepop might profitably be studied as objects of a derived category, where the derived functor construction extracts invariant information from equivalence classes of resolutions. In the Spherepop context, the relevant resolutions are the various syntactic descriptions of a single event graph, and the derived invariant is the normal form. The normalization functor of Section 8 plays the role of the derived functor, extracting the canonical representative from a class of equivalent descriptions.

Large histories may also be assembled from smaller components defined on overlapping subgraphs of the event graph. If $\{H_i\}$ is a collection of local histories defined on

overlapping regions and these histories agree on their pairwise intersections—that is, the restrictions of H_i and H_j to the subgraph $G_i \cap G_j$ are equal—then a global history may be assembled by gluing. This gluing condition is precisely the condition that appears in sheaf theory when local sections over open sets of a topological space are assembled into a global section. Spherepop histories therefore exhibit the structure of a sheaf over the underlying causal graph, where the "open sets" are the hereditary sub-DAGs and the "sections" are compatible local histories.

The sheaf-like structure has a natural operational interpretation. In a distributed computation, different processors may have direct knowledge only of the events that occurred within their local scope. The global history of the computation is assembled from these local records by verifying consistency on the overlaps and then gluing compatible pieces together. The Spherepop sheaf structure captures exactly this assembly process, and the normal form provides a canonical global section that encodes the complete causal history of the computation.

From this perspective, Spherepop may be embedded into a broader framework of derived causal categories in which objects correspond to terminal nodes of event diagrams, morphisms correspond to irreversible processes, 2-morphisms represent rewriting equivalences between histories, and normalization extracts canonical representatives from these equivalence classes. The present essay has laid the conceptual and formal foundations on which this programme of technical development can proceed.

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